INTRODUCTION TO NETWORK SYSTEMS · CLASS 4 (RECORDED VIDEO)

The wire, the glass and the air

Signals, line coding, transmission media and how fast a channel can possibly go

Class 4 of 5 · Layer 1, the physical layer

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HOW THIS VIDEO WORKS

Four pause points

I stop, you work, then I show the answer

Two formulas

Nyquist and Shannon — both are exam material

Class 5 is live

the cables, the crimp and the flashlight happen there

Paper and a pen, not a phone. You will draw signals today.

Where we are

Layer 2 is done — today we reach the bottom of the stack



done

Class 1

LIVE L2

Recap of L7–3, link layer, MAC, ARP



done

Class 2

VIDEO L2

Framing and error detection



done

Class 3

ONLINE L2

Multiple access, Ethernet, switches, VLANs, Wi-Fi



today

Class 4

VIDEO L1

Signals, line coding, media, bandwidth and capacity



next

Class 5

LIVE L1

Modulation, multiplexing, the cables in your hands, wrap-up

Still open from Class 3: your Wi-Fi box says 1200 Mbps — why do you never see it? By the end of Part 4 you can work the number out yourself.

WARM-UP · PAUSE AND ANSWER · FROM CLASS 3

What do you remember?

Q1 **A switch gets a frame for a destination it has never seen. It…**

- A drops the frame
- B floods it
- C sends an ARP request
- D sends it to the router

Q2 **Which device splits broadcast domains?**

- A a hub
- B a switch
- C a router
- D a repeater

Q3 **Why does Wi-Fi acknowledge every frame, but Ethernet does not?**

- A Wi-Fi is newer
- B the air loses frames often
- C Ethernet has no CRC
- D ACKs make it faster

Q4 **H1 → AP → router R1. What is Address 3?**

- A the AP’s MAC
- B H1’s MAC
- C R1’s MAC
- D FF:FF:FF:FF:FF:FF

By the end of today, you can…

Six learning outcomes for Class 4

1 — Part 1

Compute transmission and propagation delay, and say which one dominates

2 — Part 1

Describe attenuation, noise and distortion, and read a loss in dB

3 — Part 1

Explain why a receiver needs a clock, and how a signal can carry its own

4 — Part 2

Encode bits in NRZ, NRZI, Manchester and 4B5B

5 — Part 3

Explain why twisted pair is twisted and how light stays in a fibre

6 — Part 4

Apply Nyquist and Shannon, and say which one a question is asking for

Outcomes 1, 2, 4 and 6 are calculations. Outcome 6 is the one that appears in almost every past paper.

Signals and the clock problem



- How a bit becomes voltage, light or radio
- Two delays, three impairments, one decibel
- Why the receiver needs a clock
- The thought experiment you will run for real in Class 5

Layer 1 is the road

Back to the parcel story from Class 1

- **L7 Application** — *what to send*
- **L6 Presentation**
- **L5 Session**
- **L4 Transport**
- **L3 Network** — *the address label*
- **L2 Data link** — *the trip sheet*
- **L1 Physical** — **THE ROAD**

Layer 1 does

- turns bits into a physical signal
- sends them one after another
- fixes how long one bit lasts
- defines the cable, connector, voltage, light or radio

Layer 1 does NOT

- know frames, MACs or IP addresses
- check for errors — that is the CRC (Class 2)
- decide who may send — that is MAC (Class 3)
- care what the bits mean

The whole of Layer 1 in one question: the sender puts a 1 on the wire — how sure can the receiver be that it reads a 1, at the right moment, and how fast can we go?

A bit has to become something physical

In networking there are only three choices



Copper

a bit becomes **a voltage**

a higher or lower voltage, or a change between them

cheap, easy to connect — but loses strength fast and picks up noise



Glass fibre

a bit becomes **light**

light on or off, or brighter and dimmer

goes furthest and fastest, immune to electrical noise — but fragile and costlier to terminate



Air

a bit becomes **a radio wave**

a change in a wave's strength, frequency or timing

no cable at all — but shared with everyone, and messy

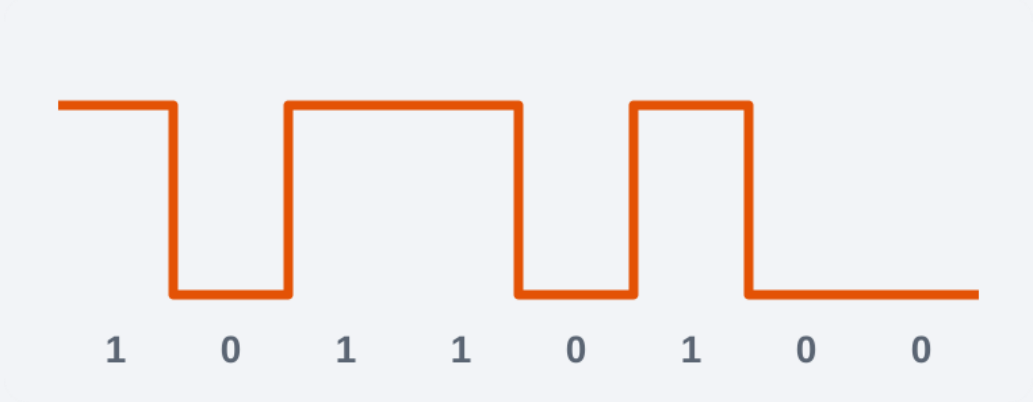
Ask: which of the three is your phone using right now? **Radio to the access point — then copper to the switch, then fibre between buildings. One message, all three media.**

Two kinds of signal

Jump between levels, or change smoothly

Digital signal

a few fixed levels, sharp jumps

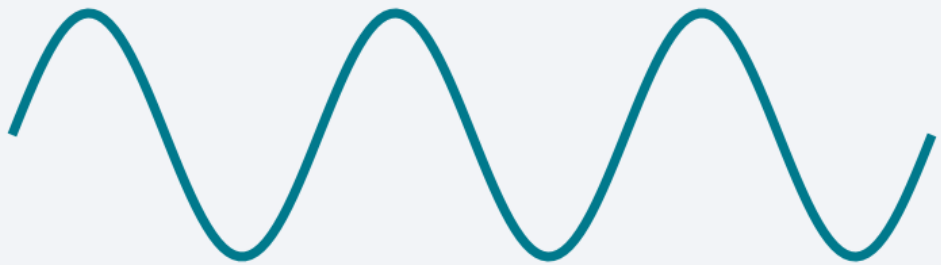


Line coding TODAY · BLOCK 2

bits sent almost directly as levels — when the medium can carry a clean square wave (copper, short distance)

Analog signal

changes smoothly and continuously



Modulation CLASS 5

bits ride on a smooth carrier wave by changing its strength, frequency or timing (radio, long lines)

Same goal, different media: line coding today, modulation next week.

How long is one bit?

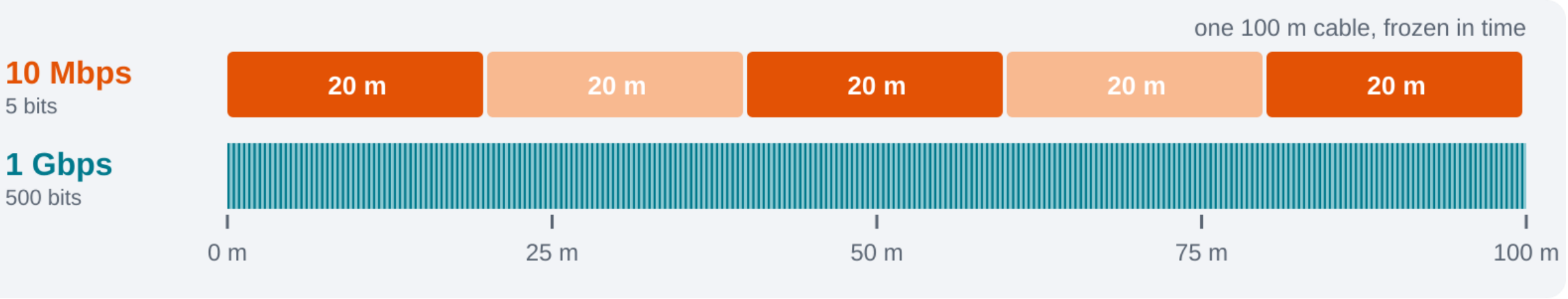
Bit time = 1 / bit rate — and on the wire, a bit has a physical length

Bit rate	Bit time	Length of one bit	Bits in flight on 100 m
10 Mbps	100 ns	**20 m**	5
100 Mbps	10 ns	**2 m**	50
1 Gbps	1 ns	**20 cm**	500
10 Gbps	0.1 ns	**2 cm**	5,000

Why a bit has a length

Signals in copper and fibre travel at about 2×10^8 m/s — two thirds of light speed in vacuum. While one bit is being sent, its front edge has already moved on:

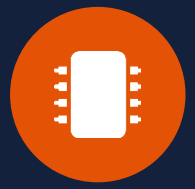
```
length = speed × bit time
```



Class 3 flashback: the 64-byte minimum frame is 512 bits × 20 m ≈ **10 km long** at 10 Mbps — long enough for a collision to get back to the sender before it finishes.

Where “100 Mbps” actually lives

It is not an abstraction. It is an object you could hold.



A sliver of quartz

A crystal oscillator on the NIC vibrates 125 million times a second. That oscillation clocks the DAC. Every tick, the line driver is told to emit the next symbol.

That is all a baud rate is.

HOW THE NUMBER IS BUILT · 100BASE-TX

125 MHz crystal	→	125 M symbols/s
4B/5B coding	→	× 0.8 b/sym
	=	100 Mbps

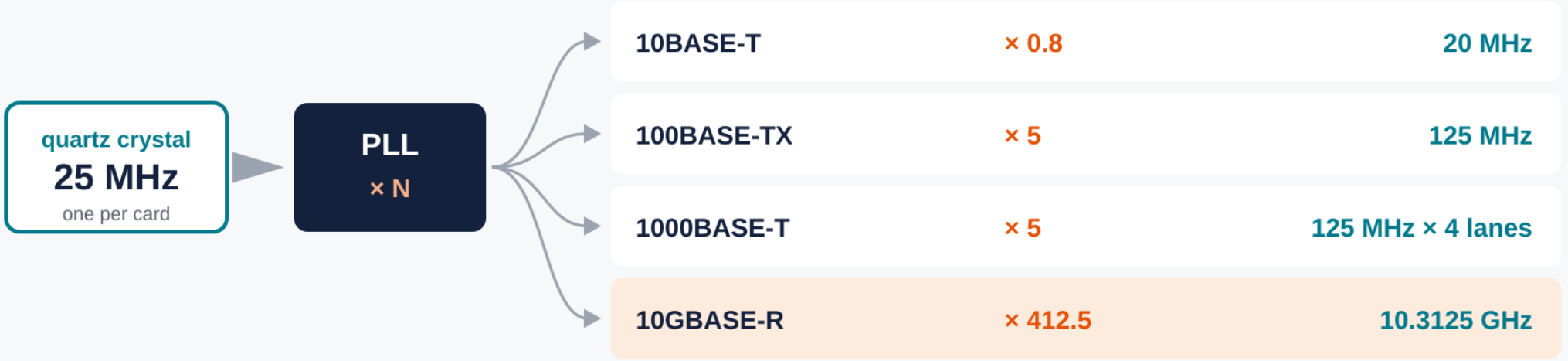
Swap the crystal for a slower one and the link rate drops — on exactly the same cable.

Classroom line: “100 Mbps” is not a property of the cable, the building, or the internet. It is the tick rate of a piece of rock in your laptop.

BUT QUARTZ STOPS AROUND 200 MHz

One crystal, every speed

You cannot buy a 10 GHz crystal. So how does a 10 Gbps PHY signal ten billion times a second? -> phase-locked loop



Real 10G cards start from a 156.25 MHz reference instead — 156.25 × 66 = 10,312.5 MHz. Same arithmetic, bigger starting point.

Why a crystal at all?

Stability. Quartz holds about ±50 ppm across temperature. A free-running oscillator would drift until the far end lost lock.

Why not a 10 GHz crystal?

Physics. Quartz tops out near 40 MHz in fundamental mode and about 200 MHz on overtones. A slab of rock cannot be made to vibrate faster.

So: the crystal sets the ACCURACY, the PLL sets the FREQUENCY. One card carries one crystal and derives every mode it supports from it — which is why 125 MHz is 100BASE-TX's number, not "a NIC's number".

ZOOM IN

What the cable actually is

Three media, three usable frequency ranges — all measured over the same 100 m.

Cat5e

100 MHz

rated to 100 MHz over 100 m

ZOOMED IN

data →



- 24 AWG copper
- 4 pairs, loosely twisted
- no separator

WHAT LIMITS IT

Leakage between pairs, plus loss that climbs with frequency.

Cat6a

500 MHz

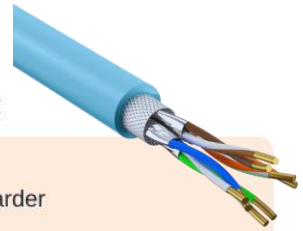
rated to 500 MHz over 100 m

ZOOMED IN

data →



- 23 AWG — thicker copper
- tighter, more varied twists
- spline separator + thick jacket



WHAT LIMITS IT

The same two problems, fought harder — not removed. Still copper.

OM3 fibre

20,000 MHz

2,000 MHz·km ÷ 100 m = 20 GHz

ZOOMED IN

data →



- 50 µm glass core
- graded index
- light, not current

WHAT LIMITS IT

Pulses spread out over distance. But nothing leaks in from next door.

All three of those numbers are frequencies, not speeds. Fibre hands you 200× the range Cat5e does — but it is still only a range. What you get out of it is decided by the transmitter.

THE PROOF

Four standards, one arithmetic

lanes × symbol rate × bits per symbol = link rate

Standard	Cable	B	Lanes	Symbol rate	Bits / symbol	Link rate
10BASE-T	Cat3	16 MHz	1	20 MBd	0.5 (Manchester)	**10 Mbps**
100BASE-TX	Cat5	100 MHz	1	125 MBd	0.8 (4B/5B, MLT-3)	**100 Mbps**
1000BASE-T	Cat5e	100 MHz	4	125 MBd	2 (PAM-5)	**1 Gbps**
10GBASE-T	Cat6a	500 MHz	4	800 MBd	3.125 (PAM-16)	**10 Gbps**

The punchline is row 3. 1000BASE-T runs on the same 100 MHz of bandwidth as 100BASE-TX, over cable barely different from Cat5 — and gets ten times the rate purely by using all four pairs instead of one and squeezing 2 bits into each symbol instead of 0.8. The copper did not change. The propagation did not change. The electronics at the two ends got cleverer.

10GBASE-T: 4 × 800 MBd × 4 b/sym = 12.8 Gbps raw; coding overhead (128-DSQ + LDPC) brings the payload to 10 Gbps.

Two delays that everyone mixes up

Loading the truck is not the same as driving it

Transmission delay

Loading the truck at the dock.

$$d_{trans} = L / R$$

Time to push the whole frame onto the wire.

- `L` frame length (bits)
- `R` link rate (bits/s)

A faster link shortens it.

Propagation delay

Driving the truck to the next city.

$$d_{prop} = d / s$$

Time for one bit to travel the length of the link.

- `d` distance (m)
- `s` signal speed ≈ 2×10^8 m/s

No upgrade changes it — only distance does.

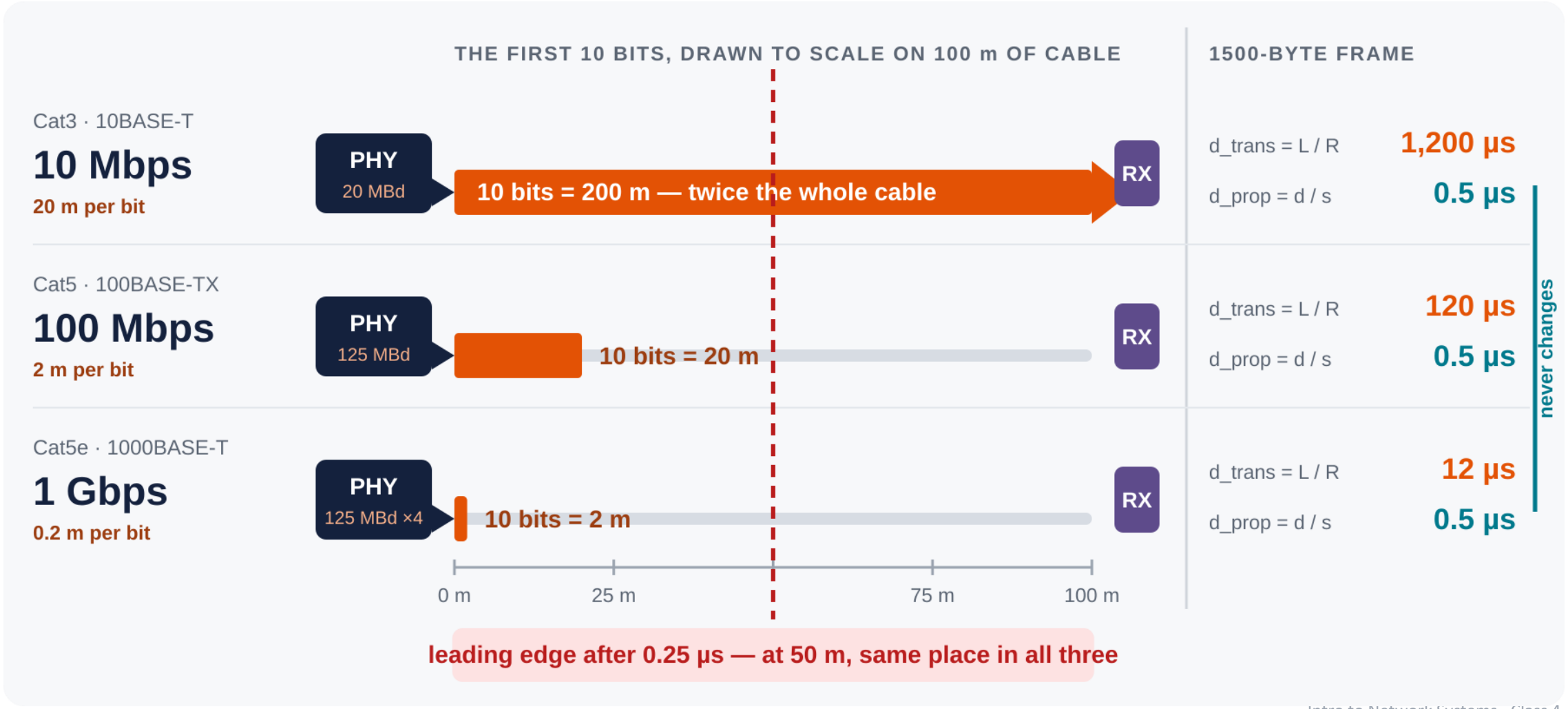
Time until the whole frame has arrived = `L / R + d / s` *the last bit leaves after L/R, then still has to make the trip*

A 10 Gbps link to Chiang Mai does not make Chiang Mai any closer.

THE ONE-PAGE PICTURE

One cable, three link rates

Same copper, same 100 m, same physics. Only the transmitter changed.



Three trips, worked out

Same idea, three very different distances

Across the room

1500-byte frame · 1 Gbps · 100 m Cat6

Transmission L / R

$12{,}000 \text{ b} / 10^9 \text{ b/s}$ = **12 µs**

Propagation d / s

$100 \text{ m} / 2\times10^8 \text{ m/s}$ = **0.5 µs**

transmission dominates

Bangkok → Chiang Mai

1500-byte frame · 1 Gbps · 700 km fiber

Transmission L / R

$12{,}000 \text{ b} / 10^9 \text{ b/s}$ = **12 µs**

Propagation d / s

$700 \text{ km} / 2\times10^8 \text{ m/s}$ = **3.5 ms**

propagation dominates (≈300×)

Via a geostationary satellite

up 35,786 km, then down again

Transmission L / R

depends on the link = **—**

Propagation d / s

$71{,}572 \text{ km} / 3\times10^8 \text{ m/s}$ = **≈ 239 ms**

a quarter-second before anything else

The Class 1 parcel took about ten hours by truck to Chiang Mai. A bit takes three and a half milliseconds — and no faster link can shorten that part.

PAUSE POINT 1 · ABOUT 3 MINUTES

Which delay dominates?

Pause here. Transmission (L/R) or propagation (d/s)? Estimate — no calculator needed.

A

1500-byte frame

1 Gbps link, 100 m cable

transmission / propagation ?

B

1500-byte frame

10 Gbps fibre, Bangkok → Chiang Mai (700 km)

transmission / propagation ?

C

64-byte frame

50 Mbps satellite link, 36,000 km up

transmission / propagation ?

D

a 1 GB file

100 Mbps link, 100 m cable

transmission / propagation ?

Write down all four before you restart. Look for the pattern: which kind of link, and which kind of data, makes each delay the big one?

PAUSE POINT 1 · ABOUT 3 MINUTES

Which delay dominates?

Pause here. Transmission (L/R) or propagation (d/s)? Estimate — no calculator needed.

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1500-byte frame

1 Gbps link, 100 m cable



transmission / propagation ?
12 µs / 0.5 µs

B

1500-byte frame

10 Gbps fibre, Bangkok → Chiang Mai (700 km)



transmission / propagation ?
1.2 µs / 3.5 ms

C

64-byte frame

50 Mbps satellite link, 36,000 km up



transmission / propagation ?
10.24 µs / 240 ms

D

a 1 GB file

100 Mbps link, 100 m cable



transmission / propagation ?
80 seconds / 0.5 µs

Write down all four before you restart. Look for the pattern: which kind of link, and which kind of data, makes each delay the big one?

Nothing arrives exactly as it was sent

Three things damage every signal on the way

Attenuation

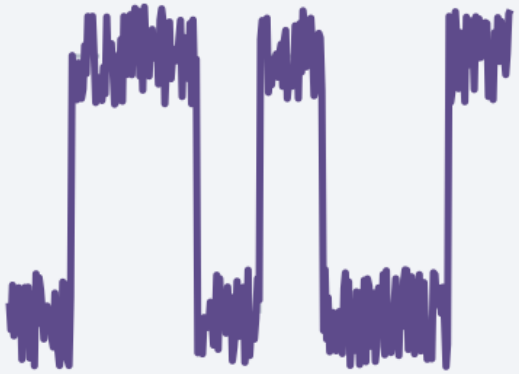


dashed = what was sent

weaker with distance

copper turns it to heat, glass to scattering, air to spreading out

Noise

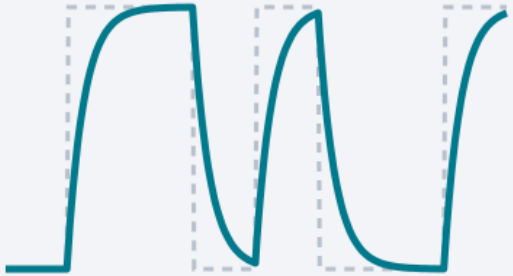


dashed = what was sent

unwanted energy added

other cables, motors, radios — and random electron motion

Distortion

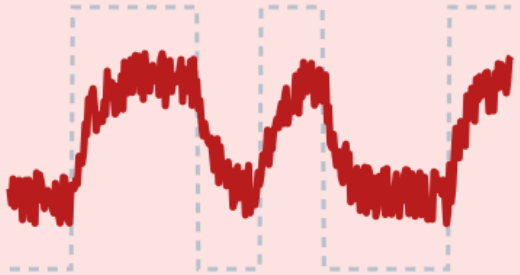


dashed = what was sent

edges get smeared

neighbouring bits start to blur into each other

All three at once



dashed = what was sent

what a receiver really gets

and it must still decide: 1 or 0?

The receiver's whole job: look at the last picture and decide, bit by bit, one or zero. Every technique today exists to make that decision easier.

Decibels in one slide

You will see dB on every cable and Wi-Fi datasheet

Start with 1000 mW. How much survives a loss of…

3 dB	500 mW	*half*
10 dB	100 mW	*a tenth*
20 dB	10 mW	*a hundredth*
30 dB	1 mW	*a thousandth*

$$dB = 10 \log_{10} (P_{in} / P_{out})$$

The useful property

Losses in dB simply add up — no multiplying.

```
cable       10 dB
connector    3 dB
total       13 dB
```

Ask

A Wi-Fi signal passes through two walls that cost 10 dB each. What fraction of the power survives?

20 dB → one hundredth

Remember four rows. 3 dB = half is the one engineers use most.

Same loss, a thousand times the distance

Why fibre runs between buildings and copper runs to your desk

Cat6 copper

at 100 MHz



≈ 20 dB over 100 m

about 1% of the power arrives

Single-mode fibre

≈ 0.2 dB per km



≈ 20 dB over 100 km

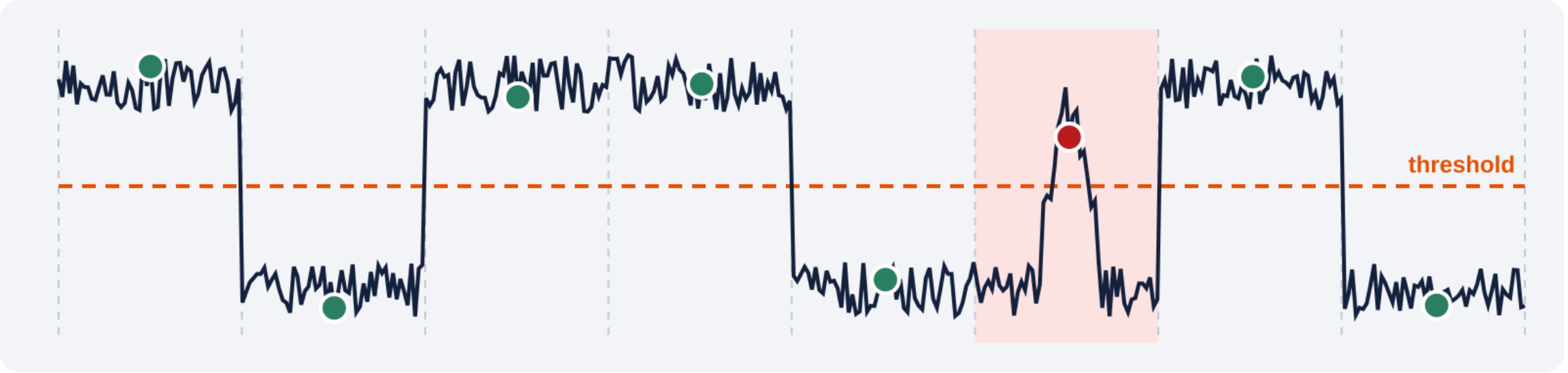
the same 1% — a thousand times further

bars drawn to scale — the copper bar is 100 m

This one fact explains most of a campus network: fibre between buildings, copper for the last 100 m. And copper loss grows with frequency — so faster copper needs better cable (Block 3).

How noise becomes a bit error

The receiver compares each bit with a threshold — in the middle of the bit



sent	1	0	1	1	0	0	1	0
read	1	0	1	1	0	1	1	0

A noise spike pushed a 0 above the threshold: a bit error. Exactly what the CRC catches (Class 2). To fight it: a stronger signal compared with the noise — the SNR, star of Class 5 — or keep the noise out (Block 3).

And notice: the receiver sampled the middle of every bit. How did it know where the middle was?

Why the receiver needs a clock

It samples each bit once — so it must tick at exactly the sender's rate



Sent: 1 0 0 0 0 0 0 0 0 1 *(eight 0s)*

Read: 1 0 0 0 0 0 0 0 1 *(seven 0s)*

Why was the first 1 → 0 safe?

Because every change in the signal lets the receiver correct its clock. A long run with no changes gives it nothing to correct against.

Long runs of identical bits are the enemy. Keep that sentence — Block 2 is built on it. (Drift exaggerated here so you can see it.)

PAUSE POINT 2 · ABOUT 4 MINUTES

The Flashlight Link

A one-way link with no way to talk back. In Class 5 we run this for real, in the room.

The link

ON = 1

OFF = 0

- one torch at the front
- seventy people writing down bits
- nothing goes back the other way

1 **No agreement**

The sender picks their own timing. Nobody has said how long a bit lasts.

2 **A shared clock**

A steady beat is clapped out loud. The light may change only on a clap.

3 **The clock goes quiet**

Four claps to set the beat — then silence. The sender keeps time in their head.

4 **Every bit blinks**

No claps at all. A 1 is OFF → ON inside one beat; a 0 is ON → OFF.

Pause here. For each round, write down whether the receiver gets the bits right — and if not, exactly what goes wrong. Then restart.

ANSWERS · PAUSE POINT 2

What each round taught

Four rounds, four separate lessons — and the last one has a famous name.

1 No agreement

agreed bit rate

The receiver must know how long a bit lasts — “one long ON” and “two ONs” look identical.

2 A shared clock

clock wire

A separate clock fixes everything — if you can afford a second channel just for timing.

3 The clock goes quiet

clock drift

Your rhythm drifted. A long run of identical bits gave you nothing to correct against.

4 Every bit blinks

self-clocking

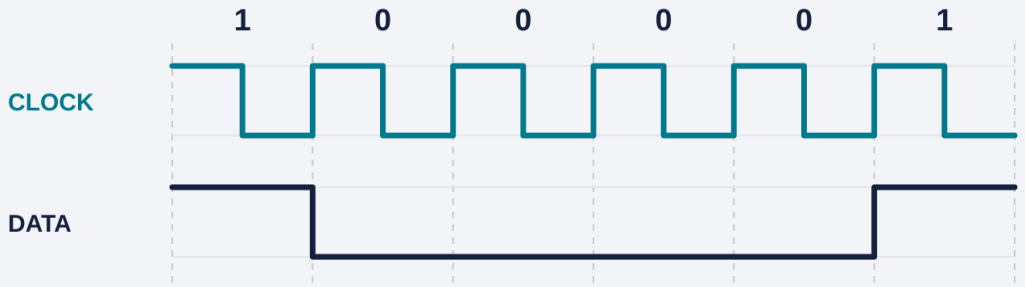
A change inside every bit lets the receiver recover the clock from the data itself — at twice the effort.

Round 4 has a name: Manchester encoding — the line code of the original 10 Mbps Ethernet. And the Class 2 preamble is the same trick: a steady run of changes to lock onto.

Two ways to share a clock

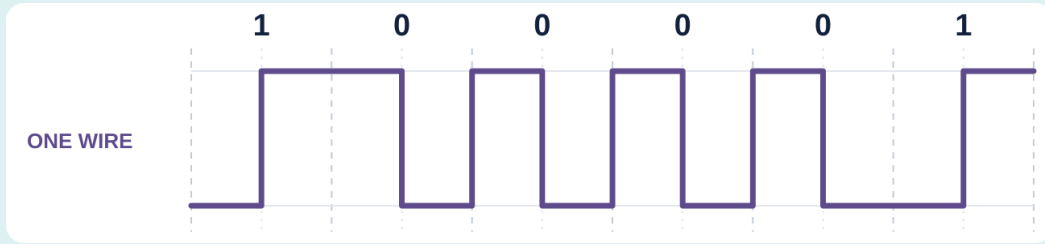
Send it separately — or hide it inside the data

A separate clock wire



- works perfectly — round 2 of the flashlight
- costs an extra wire or channel
- over distance the two arrive at slightly different times (skew)

A self-clocking signal



- the signal changes often enough to steer the clock
- one wire carries data and timing together
- what every network link does

Block 2 in one question: how do we guarantee enough changes — without paying double, as the flashlight did?

QUICK CHECK · PAUSE AND ANSWER

End of Part 1

Q1 Propagation delay depends on…

- A the bit rate
- B distance and signal speed
- C the frame size
- D the CRC

Q2 A 20 dB loss leaves what fraction of the power?

- A 1/2
- B 1/20
- C 1/100
- D 1/1000

Q3 A receiver reads a 0 as a 1 because…

- A its clock is fast
- B the cable is short
- C noise crossed the threshold
- D the CRC failed

Q4 What makes a receiver lose its clock?

- A a strong signal
- B long runs of one bit
- C short frames
- D using fibre

END OF PART 1

Round 4 worked — but the flashlight had to blink twice as often.

So here is the engineering question Part 2 answers:

how do you send a thousand 0s in a row and keep the clock — without paying double?



A good place to stop

If you are watching this in one sitting, take ten minutes here. The answer is a table of sixteen five-bit codes.

Line coding

- What a good line code needs
- NRZ, NRZI and Manchester — you draw them
- 4B5B: a flag that needs no stuffing
- Where each code lives in real Ethernet



What should a good line code do?

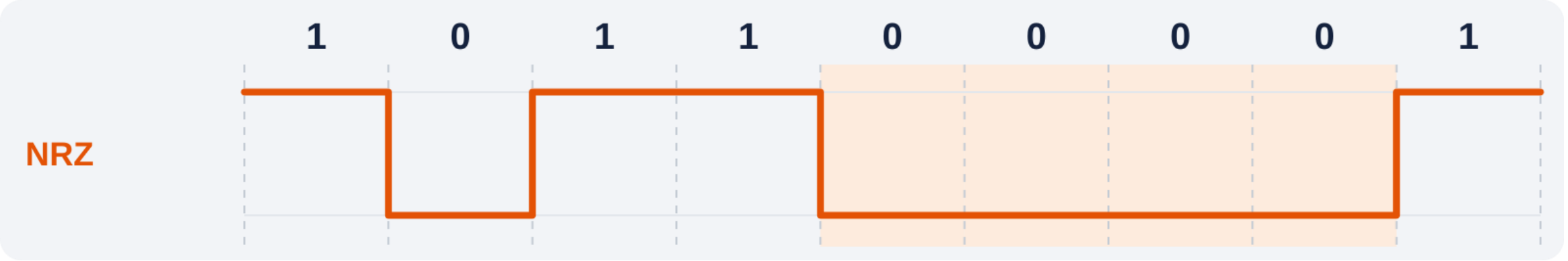
Agree the wish list before meeting any code — like the ideal MAC protocol in Class 3

1	**Enough changes**	so the receiver can keep its clock, whatever the data is
2	**Balanced (DC-free)**	about as much time high as low — no drifting average
3	**Efficient**	few signal changes per bit, because every change costs bandwidth
4	**Spare patterns**	left-over codes that can mean “start”, “end” or “something is wrong”

Spoiler: no code wins on all four. Every code you meet trades changes against bandwidth — exactly like the multiple access protocols traded fairness against efficiency.

NRZ: the simplest code

Non-return to zero — 1 is high, 0 is low, and the level holds for the whole bit



Good

At most one change per bit — as efficient as a code can be.

Bad

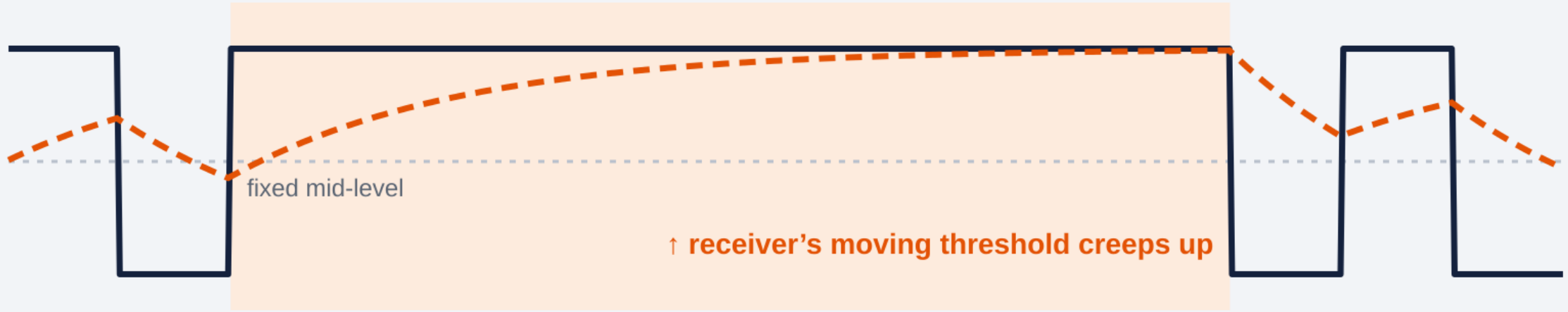
A run of 0s (or of 1s) has no changes at all. The shaded bits give the receiver nothing to keep time with.

Ask: which part of this signal would worry the receiver? **The four 0s — now imagine a thousand of them.**

Textbooks differ on whether 1 is high or low in NRZ. The idea is the same either way.

Long runs hurt twice

No changes for the clock — and a drifting average for the threshold



Problem 1 · no clock

No changes, nothing to steer the receiver's clock — round 3 of the flashlight.

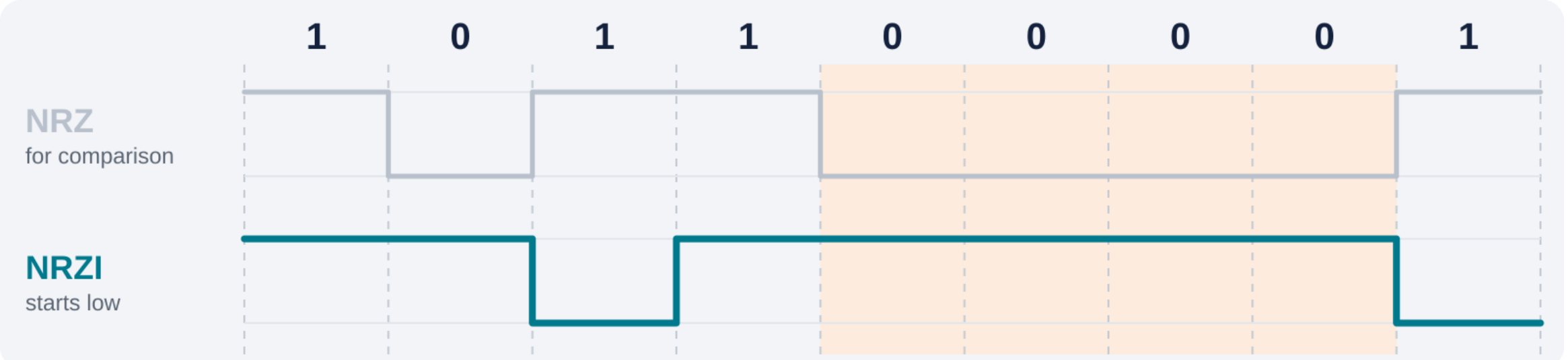
Problem 2 · baseline wander

The receiver compares with its recent average. During a long run the average creeps, until a good 1 starts to look like a 0.

A physical reason too: Ethernet ports connect through small transformers for safety, and a transformer cannot pass a level that never changes. A good code needs changes AND balance.

NRZI: put the information in the changes

1 = change level at the start of the bit · 0 = stay where you are



Runs of 1s are fixed

Every 1 creates an edge, so a thousand 1s means a thousand changes.

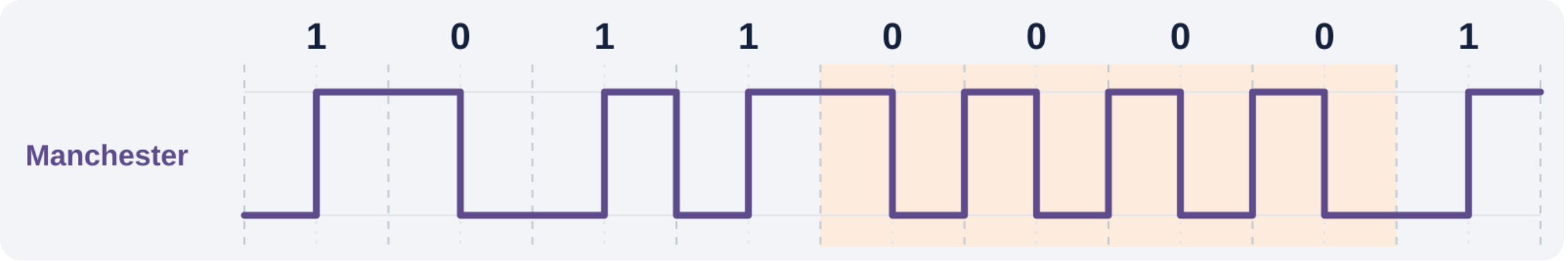
Runs of 0s are not

Still no changes in the shaded bits. We have fixed half of the problem.

A nice side effect: NRZI only cares whether the level moved, not whether it is high or low — swap the two wires and it still decodes correctly.

Manchester: a change in the middle of every bit

IEEE 802.3 convention: 0 = high → low · 1 = low → high



Always a change

Even the four 0s give four clear edges. The receiver can always recover the clock.

Perfectly balanced

Every bit is half high, half low — no drifting average, no baseline wander.

The price

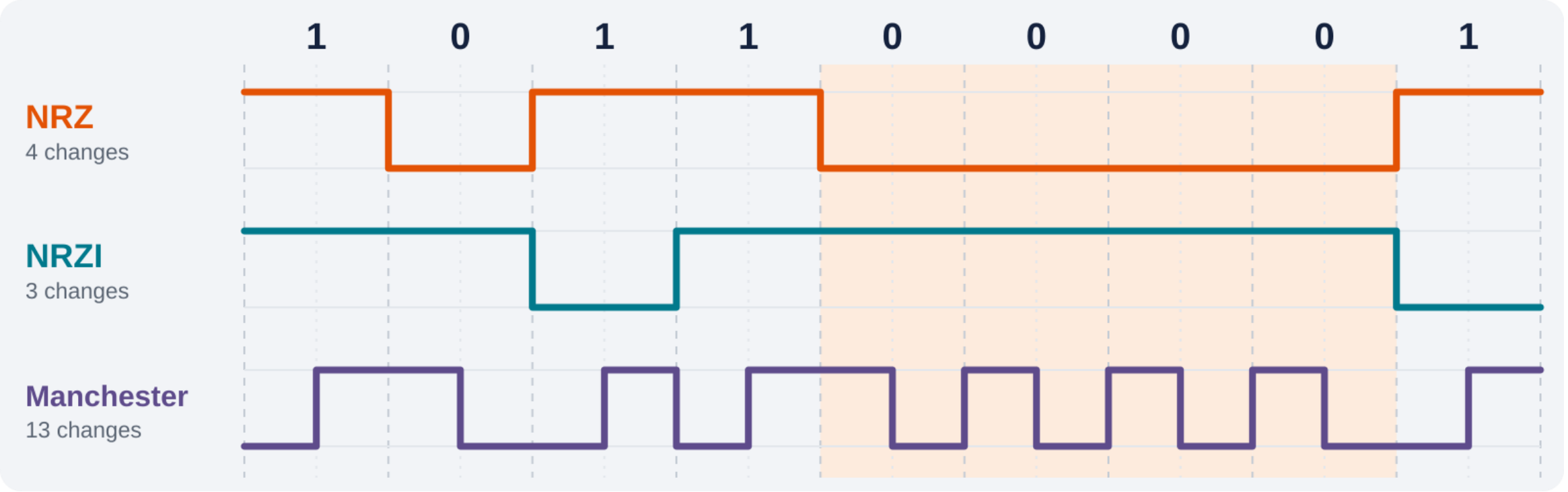
Up to two changes per bit — the signal needs about twice the bandwidth of NRZ.

You performed this with a flashlight in round 4. It is the code of 10 Mbps Ethernet — and the Class 2 preamble 1010… in Manchester is a perfectly regular signal to lock onto.

Some textbooks use the opposite convention (1 = high → low). Either is fine if you name it.

Three codes, same bits

Count the signal changes in each row

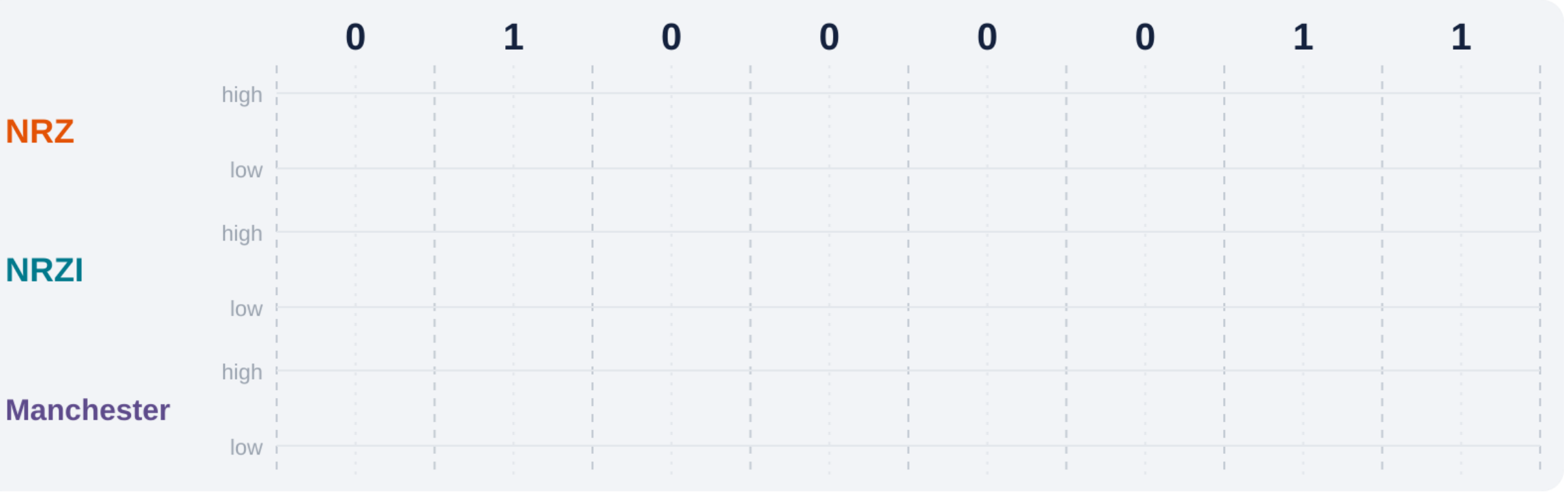


Down the page, reliability goes up — and so does the number of changes. Manchester buys its clock with bandwidth. Now draw them yourselves.

PAUSE POINT 3 · ABOUT 6 MINUTES

Draw it yourself

Pause now. Copy the grid onto paper and encode 0 1 0 0 0 0 1 1 in all three codes.



NRZ 1 = high, 0 = low

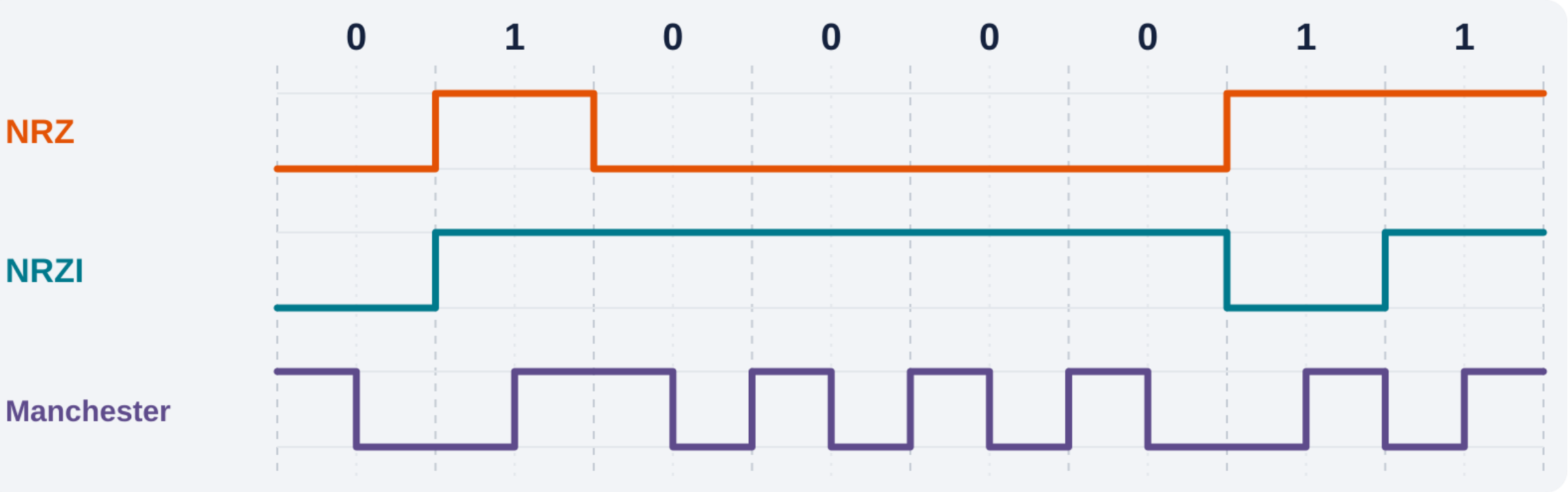
NRZI start LOW · change on every 1

Manchester 0 = high → low · 1 = low → high

ANSWERS

Check your drawings

Watch the boundaries between two equal Manchester bits — an extra edge is needed there.

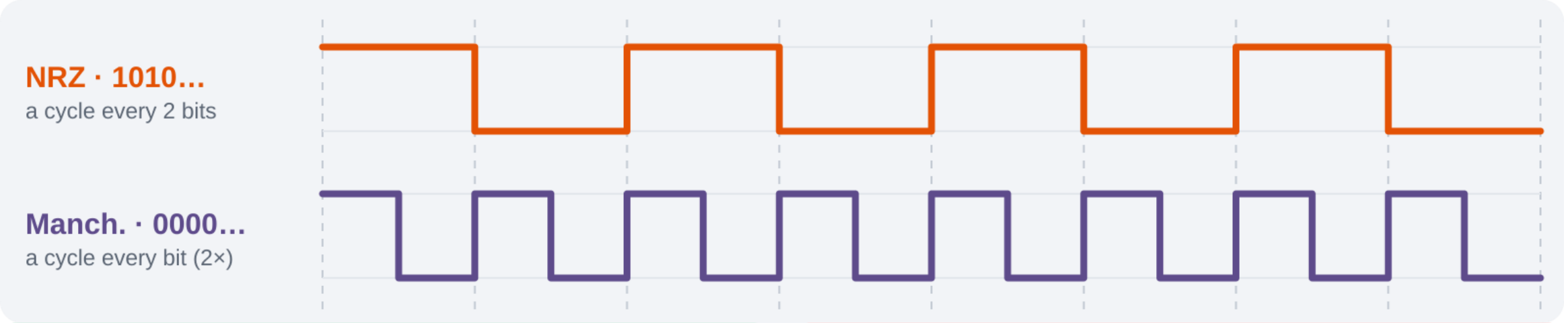


Common mistake · NRZI: changing on 0 instead of on 1

Common mistake · Manchester: forgetting the edge at the boundary between two equal bits

Manchester’s price

The fastest pattern each code can produce — and why it matters



1990 · 10 Mbps on twisted pair

Manchester doubled the frequency to 10 MHz — easy for the Category 3 cable already in buildings. No reason to be clever.

1995 · 100 Mbps

Doubling again meant 100 MHz — the entire range Category 5 is rated for. The signal would arrive at 8% strength with the neighbouring pair only 10 dB below it. No margin for a real building, and re-cabling was unthinkable.

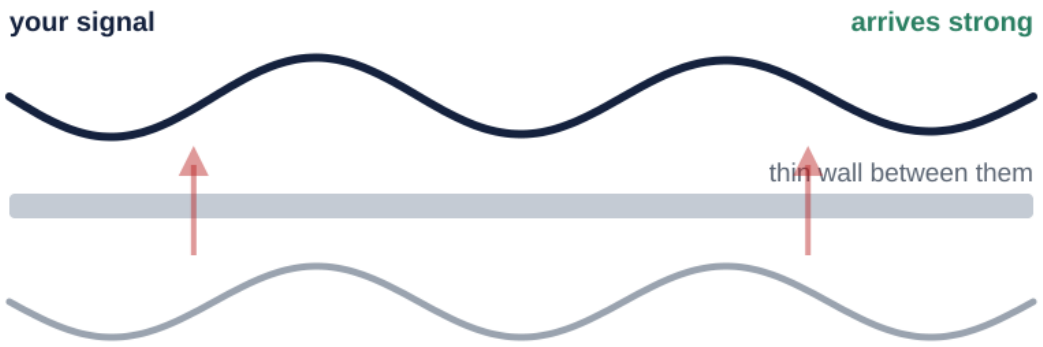
The engineering question: can we guarantee changes by adding a little redundancy, instead of doubling everything?

The answer already existed: 4B5B came from FDDI in the late 1980s and MLT-3 from its copper version (ANSI X3.263). 100BASE-TX borrowed that physical layer rather than inventing one.

Going faster costs you twice

Not one problem getting worse — two, arriving at the same time

SLOW CHANGES

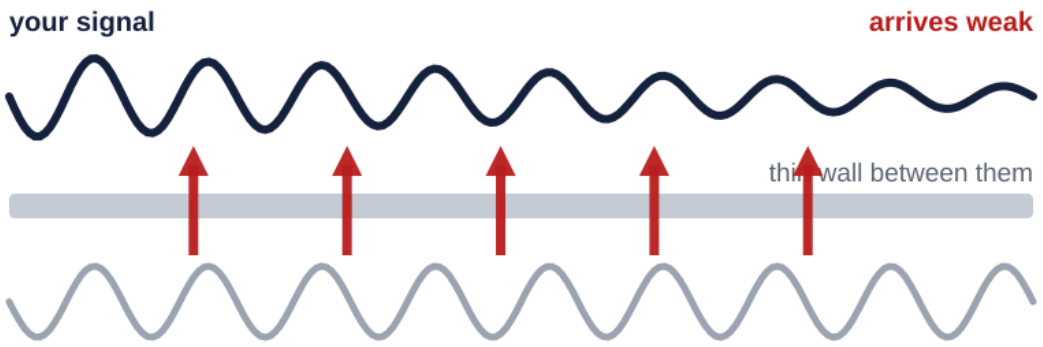


**Like leaning on a wall —
the room next door hears nothing.**

ROOM TO TELL SIGNAL FROM NOISE

plenty of room

FAST CHANGES



**Like tapping it fast —
every knock goes straight through.**

ROOM TO TELL SIGNAL FROM NOISE

almost none left

Both at once: push the signal faster and it arrives weaker, while more of the neighbour leaks in. The receiver lives on the gap between the two — and that gap closes from both sides.

4B5B: swap every 4 bits for a well-chosen 5

16 data values, 32 possible five-bit codes — keep only the 16 best

Data	Code	Data	Code
0000	**11110**	1000	**10010**
0001	**01001**	1001	**10011**
0010	**10100**	1010	**10110**
0011	**10101**	1011	**10111**
0100	**01010**	1100	**11010**
0101	**01011**	1101	**11011**
0110	**01110**	1110	**11100**
0111	**01111**	1111	**11101**

How the 16 were chosen

No code starts with more than one 0, and none ends with more than two. So wherever two codes meet, there are never more than three 0s in a row.

Then send it with NRZI

NRZI changes on every 1 — and 4B5B guarantees a 1 at least every 4 bits. The clock never starves.

The price: 25%, not 100%

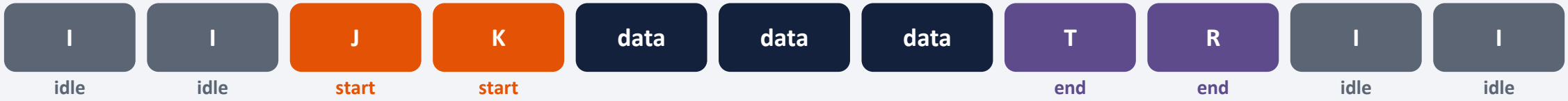
5 bits for every 4. 100 Mbps Ethernet actually sends 125 million symbols per second.

Manchester guarantees a change every bit, for 100% extra. 4B5B guarantees one every few bits, for 25%. Good enough is cheaper than perfect.

The 16 spare codes get jobs

Patterns that can never appear in the data — so they can mean something else

on the wire, all the time:



Name	Code	Meaning
I	11111	idle — sent continuously between frames
J K	11000 10001	start of a frame
T R	01101 00111	end of a frame
others	—	never valid: seeing one means an error

Class 12 flashback

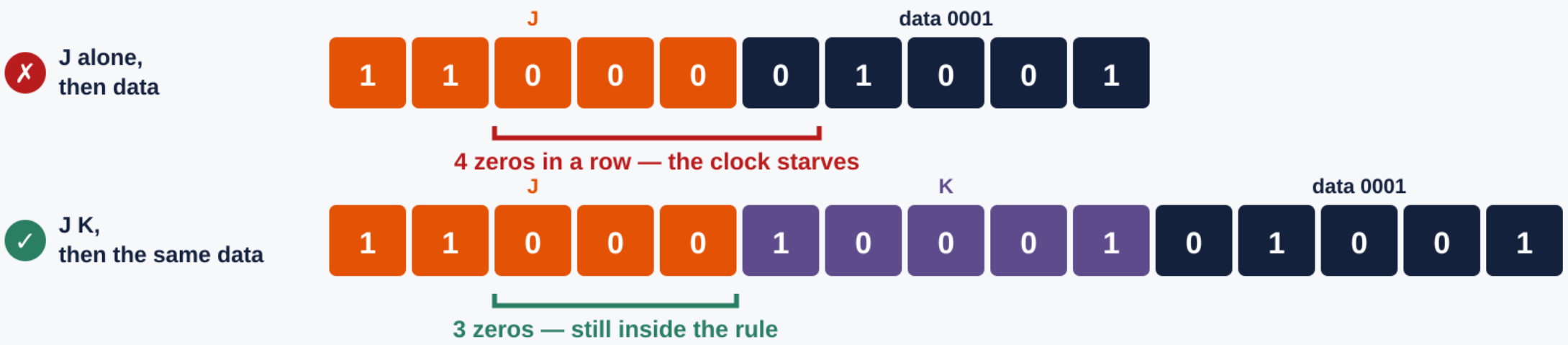
Byte and bit stuffing were needed because a flag could appear inside the data.
J K can never appear inside the data — the data only uses the other 16 codes.

A flag that needs no stuffing, because it lives outside the data alphabet.

And because idle keeps the line busy, the receiver never loses its clock — on 100 Mbps Ethernet the preamble is mostly kept for compatibility with the original frame.

Why J K, and not just J?

The second symbol is not there to be recognised — it is there to make the join legal



J alone is not even unique

Before it finds the frame the receiver is reading BITS, not symbols — there are no boundaries yet. The pattern 11000 does appear inside ordinary data, one bit off a boundary:

`11100|01001  →  1·11000·1001`

J K is exactly one byte slot

4B5B expands 4 data bits into 5 line bits — so one byte (8 bits) becomes 10 bits on the wire. J K is 10 line bits: exactly one byte's worth, so the receiver stays on the byte grid.

`8 data bits  →  10 line bits  =  one byte slot`

The same trick at the other end: R starts with two zeros, so data ending in "00" would give four again — T sits in between. In both delimiters one symbol carries the signature and the other makes the join legal.

PAUSE POINT 4 · ABOUT 3 MINUTES

4B5B by hand

Pause here and use the table on the right.

Q1 **Encode the data**

`0011 1010` → _____

Q2 **Decode the received bits**

`11110 11101` → _____

Q3 **The receiver sees this. What does it mean?**

`11000 10001` → _____

Data	Code	Data	Code
0000	**11110**	1000	**10010**
0001	**01001**	1001	**10011**
0010	**10100**	1010	**10110**
0011	**10101**	1011	**10111**
0100	**01010**	1100	**11010**
0101	**01011**	1101	**11011**
0110	**01110**	1110	**11100**
0111	**01111**	1111	**11101**

`I 11111 · J 11000 · K 10001 · T 01101 · R 00111`

ANSWERS · PAUSE POINT 4

4B5B by hand

Follow-up: could the data ever produce 11000 10001? No — those codes are not in the data table.

Q1 **Encode the data**

0011 1010 → 10101 10110

Q2 **Decode the received bits**

11110 11101 → 0000 1111

Q3 **The receiver sees this. What does it mean?**

11000 10001 → J K — the start of a frame

Data	Code	Data	Code
0000	11110	1000	10010
0001	01001	1001	10011
0010	10100	1010	10110
0011	10101	1011	10111
0100	01010	1100	11010
0101	01011	1101	11011
0110	01110	1110	11100
0111	01111	1111	11101

I 11111 · J 11000 · K 10001 · T 01101 · R 00111

Where each idea lives in real Ethernet

Do not memorise the table — watch the last column fall

Standard	Year	Line code	The idea	Extra bits
`10BASE5 / 10BASE-T`	1983	**Manchester**	a change in every bit	**100%**
`100BASE-TX`	1995	**4B5B + MLT-3**	4B5B, then a 3-level signal to lower the frequency	**25%**
`1000BASE-X (fibre)`	1998	**8b/10b**	the same idea as 4B5B, bigger blocks	**25%**
`1000BASE-T (copper)`	1999	**PAM-5, 4 pairs**	several bits per symbol, both directions at once	**Class 15**
`10GBASE-R`	2002	**64b/66b**	a tiny 2-bit header on every 64 bits	**3%**

The overhead fell from 100% to 3% as engineers found cheaper ways to guarantee changes. And gigabit copper sends more than one bit per symbol — hold that thought, because Part 4 says exactly how far you can push it.

QUICK CHECK · PAUSE AND ANSWER

End of Part 2

Q1 Which code changes in the middle of every bit?

- A NRZ
- B NRZI
- C Manchester
- D 4B5B

Q2 Which runs can still starve an NRZI receiver?

- A runs of 1s
- B runs of 0s
- C both
- D neither

Q3 How much extra does 4B5B send?

- A 5%
- B 25%
- C 50%
- D 100%

Q4 A receiver sees 11000 10001. This means…

- A data 0 and 1
- B an error
- C start of a frame
- D idle

END OF PART 2

A strand of glass thinner than a hair carries more than all the copper in this building.

Light travels in straight lines. The cable does not.

So how does the light stay inside when the fibre bends?



The second good place to stop

Part 3 is copper, glass and air — the one with pictures rather than arithmetic.

Copper, glass and air



- Twisted pair, and why it is twisted
- A live crimp — RJ45 and T568B
- Fibre: light that cannot escape
- Radio and satellite: the unguided media

Guided and unguided media

Kurose’s two families of physical media

Guided

the signal stays inside something solid

- **Twisted pair**
- **Coaxial cable**
- **Optical fibre**

Private and predictable: the signal goes where the cable goes.

Unguided

the signal spreads through open space

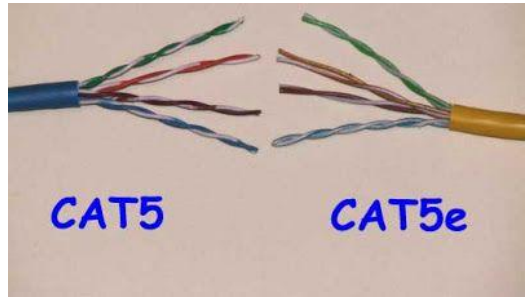
- **Terrestrial radio**
- **Satellite**

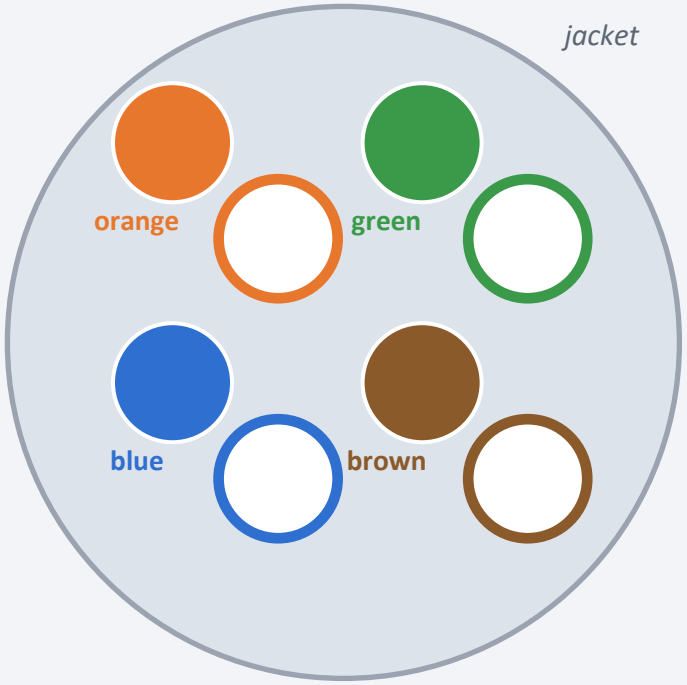
Shared by everyone in range — which is exactly why Class 3 needed CSMA/CA.

We take them in that order — starting with the one in the wall next to you.

Twisted pair: four pairs in one jacket

The cable in the wall of this room





each colour + its white-striped partner = one pair

UTP — unshielded

What is in the walls here: cheap, thin, flexible.

STP / FTP — shielded

Adds foil around the pairs. Used near heavy electrical noise, like factories.

Different twist rates

Each pair is twisted at its own rate. Two slides from now you will see why.

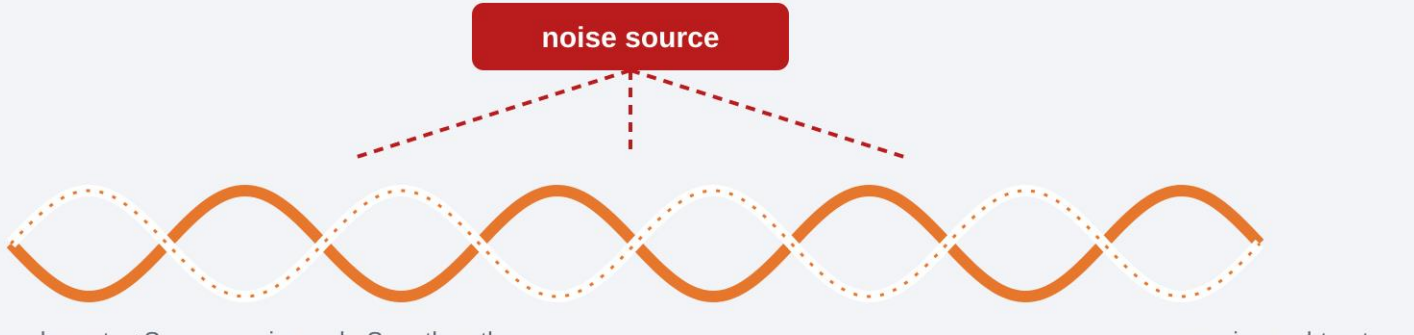
100 m, always

Every copper Ethernet standard stops at about 100 m — attenuation and noise.

8 wires in 4 pairs — orange, green, blue, brown, each with a white-striped partner.

Why twist? Noise cancels itself

Differential signalling: the receiver reads the difference between the two wires




```
wire A    S + n
wire B   −S + n
─────────────────
A − B     2S
```

The noise n cancels. The signal doubles.

Why it works

The twisting keeps swapping which wire is nearer the noise, so both wires pick up the same n.

Why different twist rates

Neighbouring pairs with different twists do not pick up each other's signals — that leakage is crosstalk.

The twist is not for strength or tidiness. It is a noise-cancelling device made out of geometry — which is why untwisting too much when you crimp a plug is a real fault.

Cable categories

The key number is the highest frequency the cable is built for

Category	Built for	Speed	Distance	Where you meet it
Cat5e	100 MHz	**1 Gbps**	100 m	*most existing buildings*
Cat6	250 MHz	**1 Gbps (10 Gbps to 55 m)**	100 m	*new office cabling*
Cat6A	500 MHz	**10 Gbps**	100 m	*Wi-Fi 6/7 access points, servers*
Cat8	2000 MHz	**25 / 40 Gbps**	**30 m**	*inside a data-centre rack*

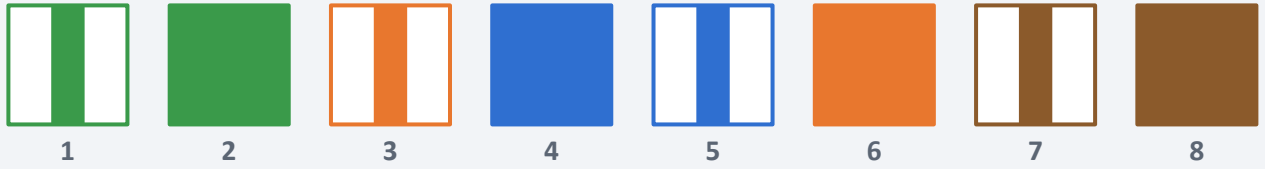
Notice the Distance column: copper holds its 100 m — or loses distance — as speed rises. Higher frequency means more loss (Block 1). Beyond the building, fibre takes over.

Tighter twists and better construction are what let each category carry higher frequencies.

The RJ45 plug and its colour order

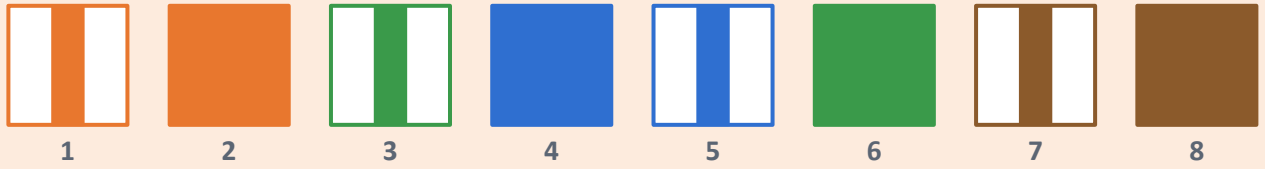
Two standards, T568A and T568B — only the orange and green pairs swap

T568A



T568B

the common choice in Thailand and most of the world



Straight-through

both ends use the same standard — PC to switch

Crossover

A on one end, B on the other — once needed for PC to PC or switch to switch

Auto-MDI-X

modern ports detect and fix it themselves — today almost any cable works

Pins actually used

10 and 100 Mbps use pins 1, 2, 3, 6. Gigabit uses all eight

What still matters today is consistency: pick one standard for a whole building and never mix them inside one cable run.

Coaxial cable

One conductor, completely surrounded by its shield



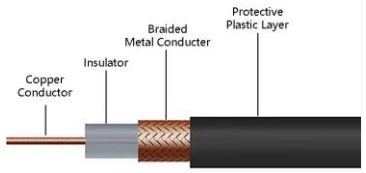


jacket

shield

insulator

centre conductor



Why it resists noise

The shield surrounds the signal on every side, so outside noise has almost no way in — and the signal has almost no way out.

Where you meet it

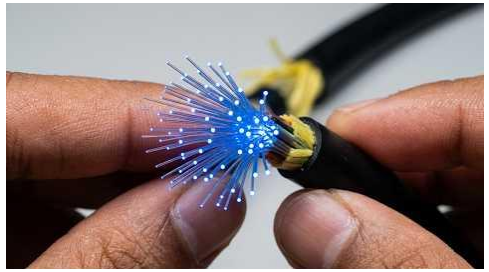
Cable TV and some cable internet services, CCTV cameras, antenna leads.

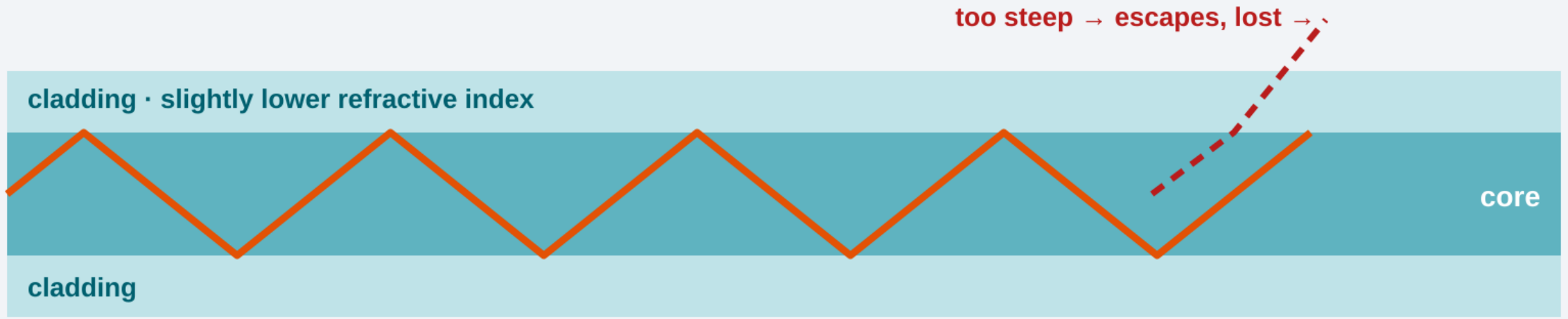
The original Ethernet

The shared bus in Class 3 was a thick yellow coaxial cable, every machine tapped onto it — 10BASE5, 500 m per segment.

How light stays inside a fibre

Total internal reflection — the answer to the break puzzle





shallow angle → reflected back into the core, every time

Total internal reflection

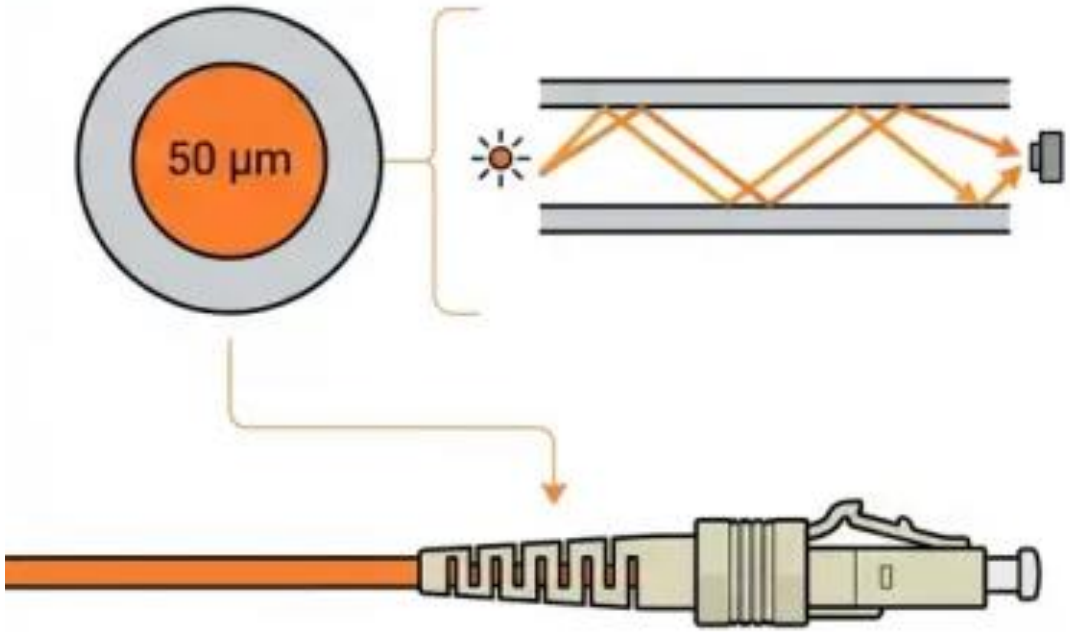
Light meeting the core–cladding boundary at a shallow angle is reflected completely. It bounces along the core, even round gentle bends.

Why fibre has a minimum bend radius

Bend it too sharply and the light meets the boundary too steeply — it escapes into the cladding and the signal is lost.

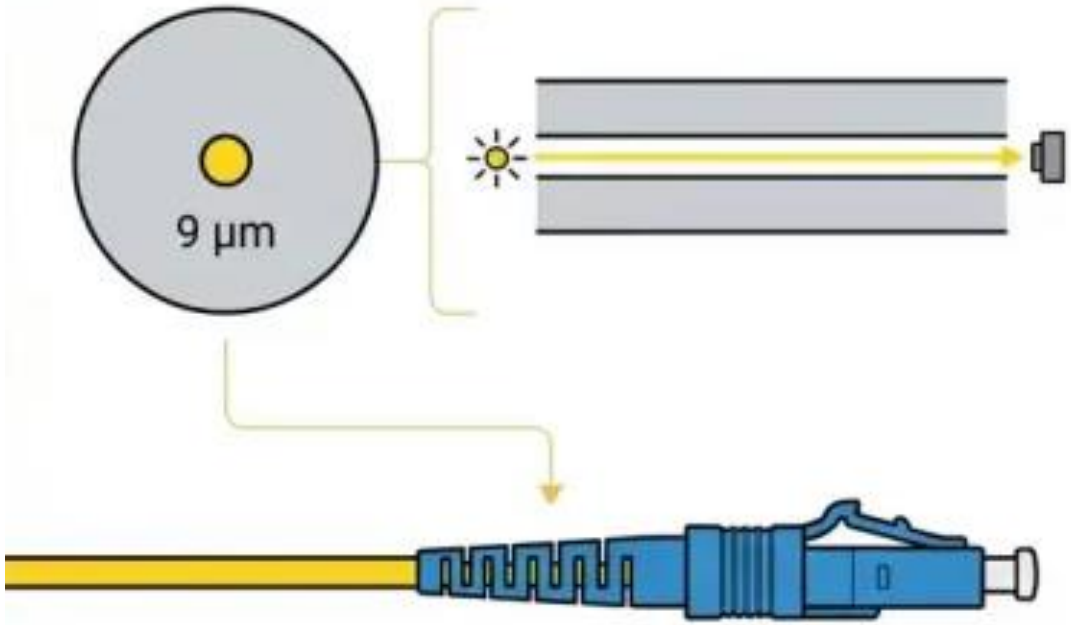
Optional demo: shine a laser pointer along a thin stream of water from a bottle — the light follows the curve. Keep it away from eyes.

Multimode Fiber



Short runs, LAN & data center

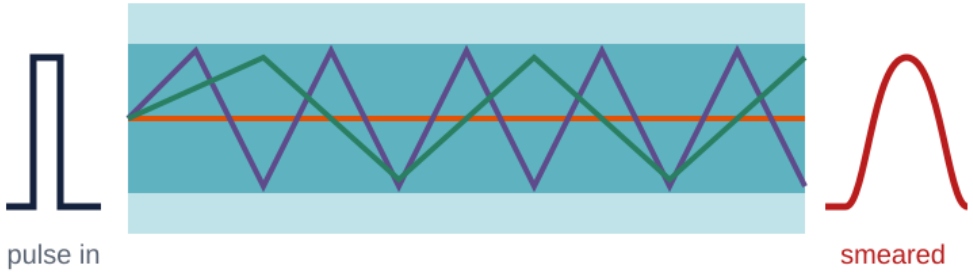
Single-Mode Fiber



Long-distance, high bandwidth links

Multimode or single-mode?

Multimode



Core 50 or 62.5 µm

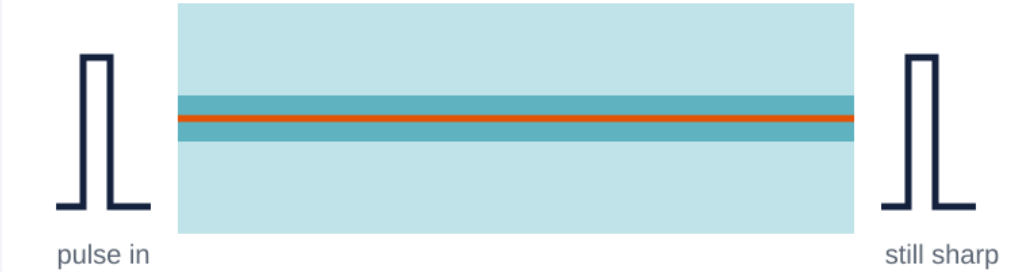
Light source LED / VCSEL, 850 nm

At 10 Gbps about 300–400 m

Jacket aqua or orange

Many paths of different lengths — pulses smear.

Single-mode



Core about 9 µm — thinner than a hair

Light source laser, 1310 / 1550 nm

At 10 Gbps 10 – 40 km and more

Jacket yellow

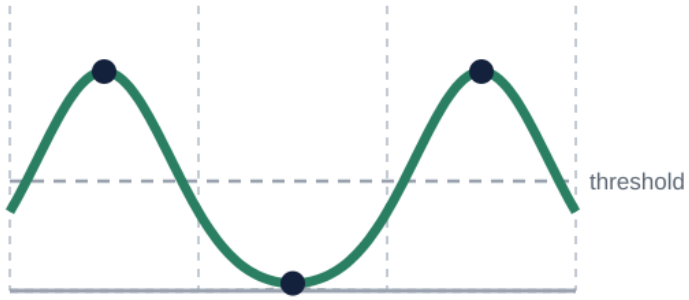
One path only — pulses stay sharp.

Rule of thumb: multimode inside a building or data centre; single-mode between buildings and across the country.

What actually runs out: the pulse spreads

The light still arrives — it just arrives smeared into the next bit's time slot

100 m
smear = 18% of a bit

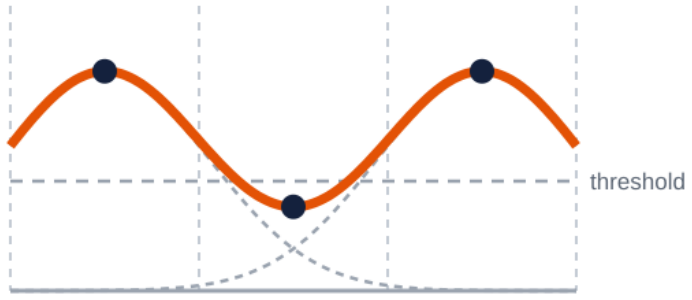


sent: 1 0 1

read: 1 0 1

read correctly

300 m
smear = 54% of a bit
the OM3 spec limit

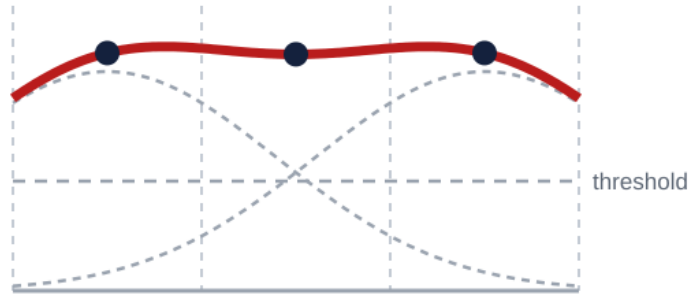


sent: 1 0 1

read: 1 0 1

still correct — only just

600 m
smear = 108% of a bit



sent: 1 0 1

read: 1 1 1

the 0 has vanished

This is intersymbol interference. Nothing ran out — the light arrived fine. It arrived spread across the slot where the next bit belonged, so the receiver could no longer tell the bits apart.

And copper does exactly the same thing: there the cable strips the high frequencies so the square edges round off, instead of light taking many paths. Different cause, identical result — and it is why every medium has a distance limit at a given speed.

Why fibre wins the backbone

And why it has not replaced the cable to your desk

Wins

Tiny loss kilometres instead of 100 m (Block 1)

Huge capacity far more than any copper cable

Immune to electrical noise light ignores motors and power cables

Carries no electricity safe between buildings, even in a lightning storm

Hard to tap unnoticed and light and thin to install

Costs

Fragile
a minimum bend radius; glass can crack

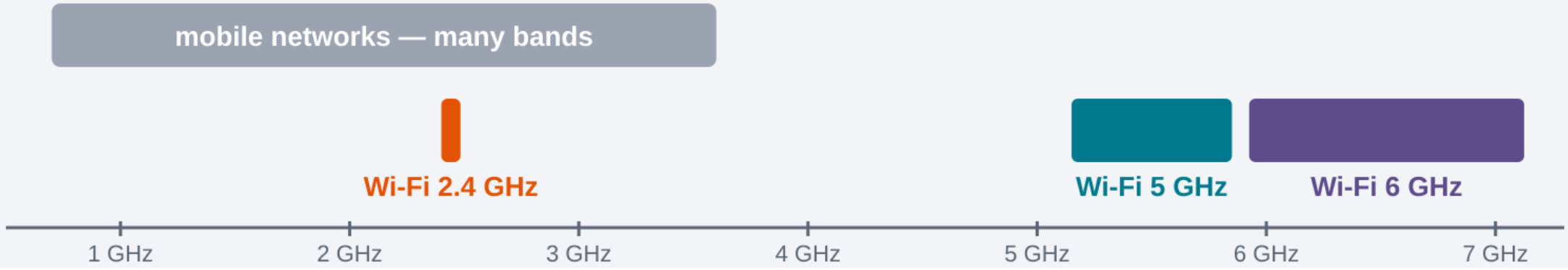
Needs skill to terminate
special tools, and a clean, polished end

Costlier ends
the optics at each end cost more than a copper port

A typical campus, including ours: single-mode fibre between buildings, copper for the last 100 m to the desk and to the access points.

Radio: no cable at all

Mobility and zero installation — at a price



Broadcast by nature

Anyone in range can hear — which is why Class 3 needed CSMA/CA, and why Wi-Fi needs encryption.

Shared with everything else

Every device on the same frequency competes — neighbours' access points, Bluetooth, even microwave ovens at 2.4 GHz.

Every medium so far kept the signal to itself. Radio shares it with the whole room — so the air does things to a signal that no cable ever does.

What the air does to a signal

Kurose’s wireless link characteristics — why Wi-Fi is not a cable



Path loss

In free space, double the distance and a quarter of the power arrives — 6 dB lost per doubling.



Obstacles

Walls, floors and people absorb signal. A concrete wall can cost 10 dB or more — and higher frequencies lose more.



Multipath

The signal bounces off walls and ceilings and arrives by several paths. The copies add up or cancel out — move your phone a few centimetres and the bars change.



Interference

Anything else on the same frequency: other access points, Bluetooth, and microwave ovens at 2.4 GHz.

All four push the same way: a weaker, noisier signal at the receiver. What the radio does about it — it slows down — is the story of Class 5.

Why Wi-Fi uses three bands

Lower frequencies reach further; higher frequencies have more room

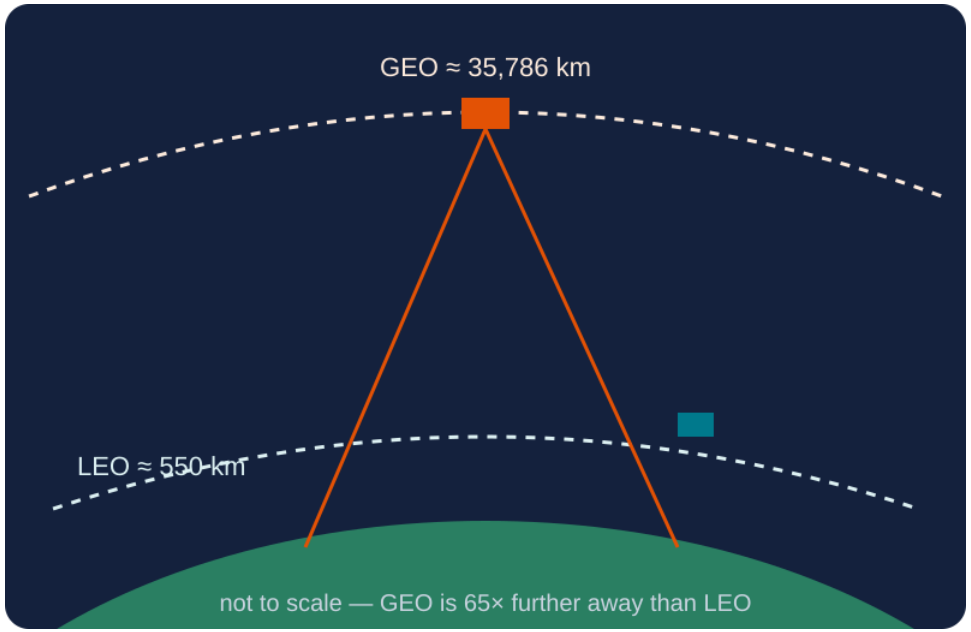
	2.4 GHz	5 GHz	6 GHz
Range and walls	best	medium	shortest
Clean channels	only 3 (1, 6, 11)	many	the most
How crowded	very	moderate	quiet — for now
Devices	everything	most	newer devices only, where permitted

Ask: a lecture hall with 70 phones — which band do you want them on? **5 or 6 GHz: many channels, and the short range is fine inside one room.**

The trade-off is physics: higher frequencies are absorbed more by walls, but there is far more spectrum up there.

Satellite: the longest link of all

Distance you can calculate, and cannot upgrade away



Geostationary (GEO)

- 35,786 km up, fixed over one point on Earth
- one satellite covers about a third of the planet
- up and down ≈ 239 ms one way — before anything else
- the dish never has to move

Low Earth orbit (LEO)

- around 550 km up
- up and down ≈ 4 ms
- each satellite crosses the sky in minutes
- needs thousands of them, and constant handover

The same lesson as Block 1: you cannot upgrade your way out of propagation delay. You can only move closer.

LEO delay shown for a satellite directly overhead; real paths are a little longer.

All the media on one page

Read it as a decision tool: distance, speed, environment

Medium	Signal	Speed	Distance	Noise	Typical use
Twisted pair	voltage	up to 10 Gbps (Cat6A)	**100 m**	good — twists	*to the desk, to the AP*
Coaxial	voltage	Gbps (cable internet)	**hundreds of m**	very good — shield	*cable TV, CCTV*
Multimode fibre	light	10 – 100+ Gbps	**a few hundred m**	immune	*inside a building, data centre*
Single-mode fibre	light	100+ Gbps	**tens of km and more**	immune	*between buildings, backbone*
Wi-Fi	radio	up to Gbps, shared	**tens of m**	poor	*anywhere you move*
Satellite	radio	up to hundreds of Mbps	**global**	weather	*remote places, ships, planes*

Two campus buildings, 800 m apart

→ *single-mode fibre*

A printer in the same room

→ *twisted pair*

An outdoor event, no cabling

→ *radio*

Why you never see 1200 Mbps

The question from Class 3 — three classes, one number

The number on the box — the promise

the radio’s top rate in perfect conditions: two antennas, a wide channel, a strong clean signal

Acknowledgments, backoff, beacons, headers — Class 3

a large share of the air is spent on things that are not your data

One channel, shared, one speaker at a time — Class 3

everyone on the channel takes turns, and Wi-Fi cannot send and receive at once

Distance, walls, multipath, interference — today

by the time it reaches you, the signal is weaker and noisier

Weaker signal → slower rate — next

the radio must drop to a slower, safer rate — and Part 4 shows you exactly why

Each row shrinks what you actually get. Part 3 explained row 4. Part 4, starting now, explains the last one — and gives you the formula that produces the number.

BLOCK 4 · ~35 min

How fast can it go?



- Bandwidth: hertz and bits per second are not the same thing
- Nyquist: how many symbols fit on a clean channel
- Shannon: what noise does to that number
- Two formulas, and how to tell which one a question wants

The word “bandwidth” means two things

Mixing them up is the single most common mistake in this topic

Bandwidth of a channel

measured in hertz (Hz)

The RANGE of frequencies the channel will carry. A property of the physical thing: the copper, the glass, the slice of radio spectrum.

- a voice telephone line: about 3,100 Hz
- a Wi-Fi channel: 20, 40, 80 or 160 MHz
- Cat 6 cable: certified to 250 MHz

Bandwidth in everyday speech

measured in bits per second (bit/s)

The RATE of information. What your ISP sells you. Strictly this is the data rate or throughput, not bandwidth at all.

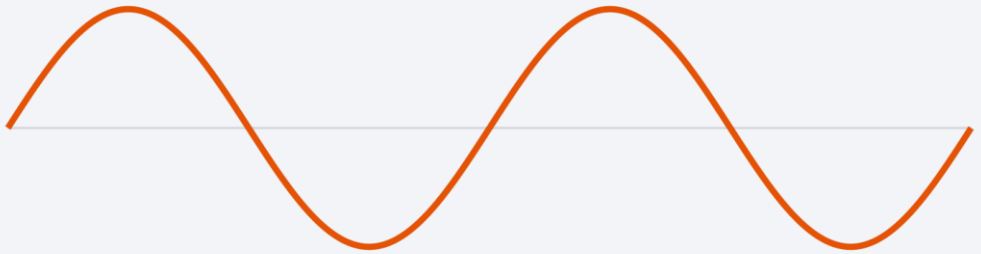
- “a 500 Mbps package”
- 1000BASE-T = 1 Gbps
- what you measure on a speed test

The rest of Part 4 is one question: given B hertz of channel, how many bits per second can you actually get out of it? Nyquist gives one answer, Shannon gives a better one.

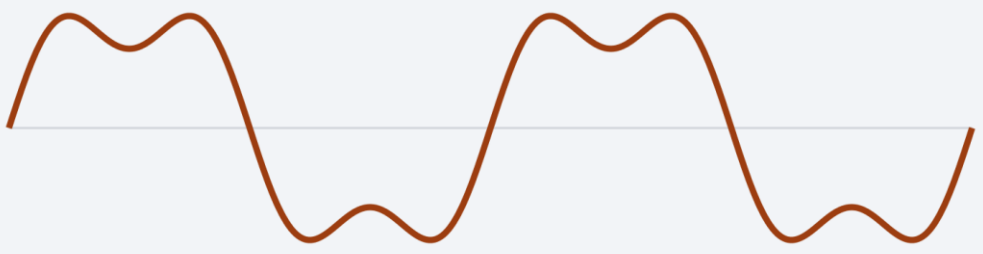
Why a channel has a bandwidth at all

A square pulse is built from many frequencies — the channel keeps only some of them

1 harmonic *just a sine wave*



3 harmonics *edges getting sharper*



9 harmonics *edges getting sharper*



49 harmonics *nearly square*



A perfectly square pulse needs infinitely many frequencies, and no real channel passes them all. Cut the high ones and the edges round off — push the rate up and neighbouring bits smear into each other. That is the physical origin of every speed limit in this part.

Nyquist: how many symbols fit

A perfect, noiseless channel of B hertz

maximum symbol rate = 2B symbols per second

You cannot squeeze more than two signal changes per hertz of bandwidth through the channel. Anything faster and the pulses merge.

But a symbol need not be one bit

If the signal can take M distinguishable levels, each symbol carries $\log_2 M$ bits.

2 levels	→ 1 bit / symbol
4 levels	→ 2 bits / symbol
16 levels	→ 4 bits / symbol
256 levels	→ 8 bits / symbol

Putting the two together

$$C = 2B \cdot \log_2 M \quad \text{bit/s}$$

- B = channel bandwidth, in hertz
- M = number of signal levels
- C = maximum bit rate, bit/s

Notice what this says: to go faster on a fixed B, add more levels. Nothing here stops you using a million of them — which should feel wrong. It is.

Nyquist, worked

A telephone line: B = 3,100 Hz

Levels M	bits/symbol = $\log_2 M$	$C = 2B \cdot \log_2 M$	Maximum bit rate
2	1	2 × 3,100 × 1	6,200 bit/s
4	2	2 × 3,100 × 2	12,400 bit/s
16	4	2 × 3,100 × 4	24,800 bit/s
256	8	2 × 3,100 × 8	49,600 bit/s
1,048,576	20	2 × 3,100 × 20	124,000 bit/s

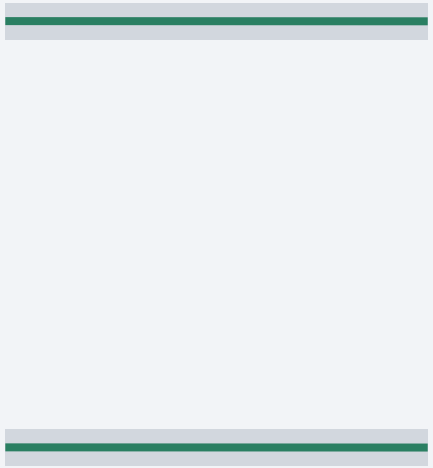
The last row is nonsense, and that is the point. A real telephone line never carried 124 kbit/s. Nyquist has no term for noise, so on paper you can always add levels. In reality the levels get closer together until noise makes them impossible to tell apart.

So Nyquist gives an upper bound that is only honest when the channel is clean. Shannon fixes exactly this.

Why not a million levels?

Same voltage range, same noise — only the number of levels changes

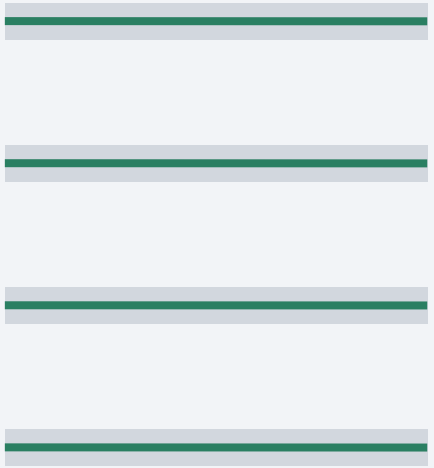
M = 2
1 bit / symbol



+0 dB

miles apart

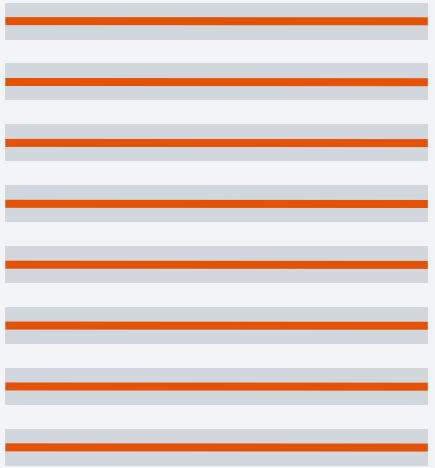
M = 4
2 bits / symbol



+9.5 dB

still clean

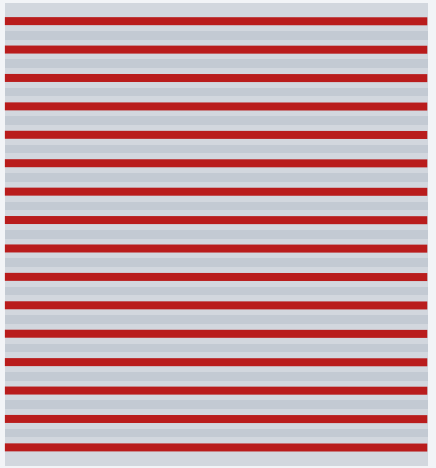
M = 8
3 bits / symbol



+16.9 dB

nearly touching

M = 16
4 bits / symbol



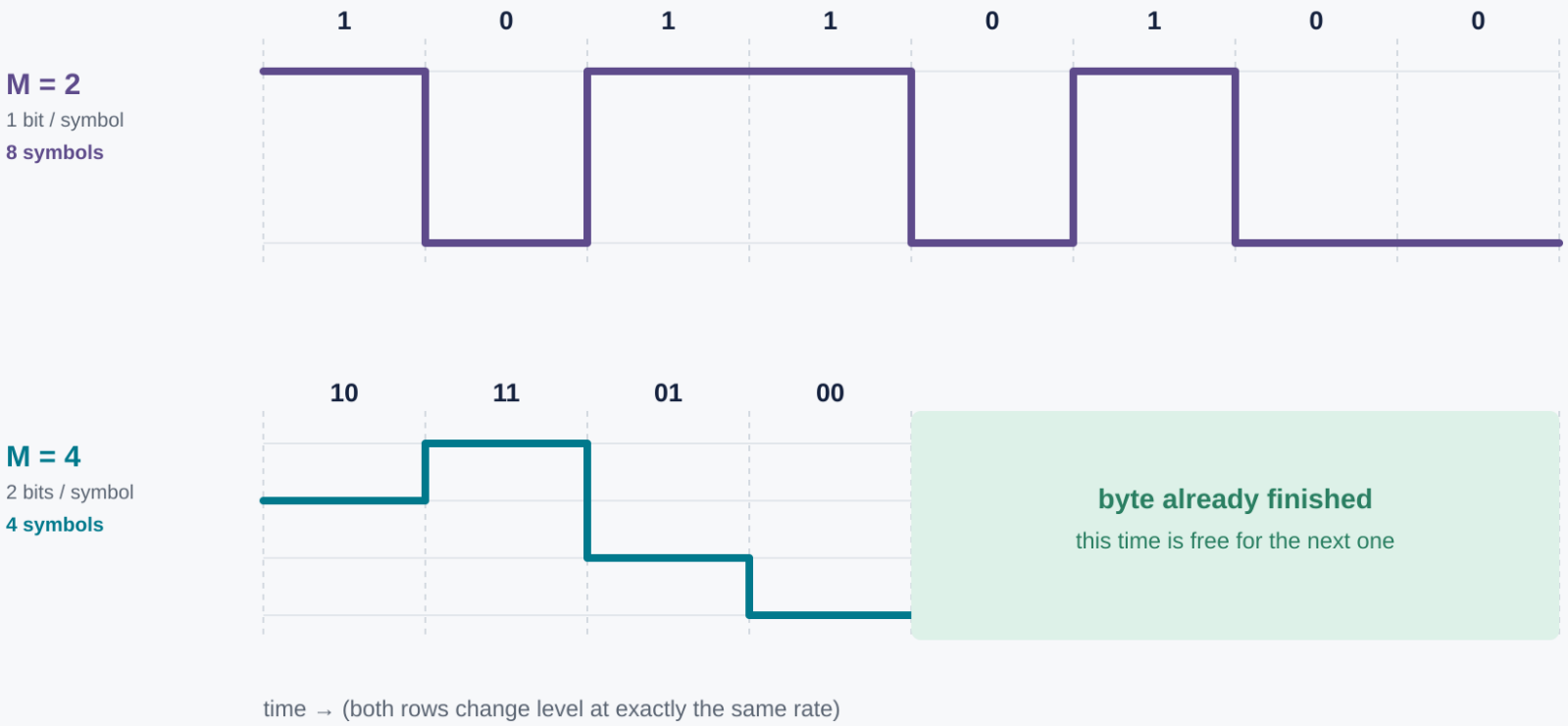
+23.5 dB

they overlap

The grey band is the noise — the same in all four. It never grew; the levels moved closer. Doubling M buys one more bit and costs about 6 dB — and Shannon turns that into a hard number.

One voltage at a time — so where do two bits fit?

The same byte 10110100 down the same wire, at the same symbol rate



The mapping for M = 4

00	→	level 0
01	→	level 1
10	→	level 2
11	→	level 3

The receiver measures ONE voltage, decides which of the four it is, and outputs TWO bits.

The two bits are not sent side by side. They are the NAME of which level to send — so the wire still carries one voltage at a time, it just says more each time it changes.

Where does the “mega” come from?

Not from M — from the symbol rate, because that is the half with time in it

$\log_2 M = 1$ bit / symbol

a RATIO

No time in it at all. It would still be 1 bit per symbol if you sent one symbol per century.

$2B = 2{,}000{,}000$ symbols/s

a RATE

This is the only half with “per second” in it — so this is the only half that can carry a kilo or a mega.

1 bit/symbol × 2,000,000 symbols/s = 2,000,000 bits/s

the word “symbol” cancels, top and bottom — exactly like units in a physics calculation

KEEP M = 2 AND CHANGE ONLY THE BANDWIDTH

B = 1 Hz	2B = 2 sym/s	× 1 bit	**2 bps**
B = 1 kHz	2B = 2,000 sym/s	× 1 bit	**2 kbps**
B = 1 MHz	2B = 2,000,000 sym/s	× 1 bit	**2 Mbps**
B = 100 MHz (Cat5)	2B = 200,000,000 sym/s	× 1 bit	**200 Mbps**

M was 2 in every row. The cable’s bandwidth supplies the “mega” — M only supplies the multiplier on top of it.

Signal-to-noise ratio, and the decibel trap

Every capacity question hides a dB → ratio conversion

SNR is a plain ratio

S / N = signal power ÷ noise power

An SNR of 1000 means the signal is a thousand times stronger than the noise. No units — it is a ratio.

But it is quoted in decibels

$SNR(dB) = 10 \cdot \log_{10}(S/N)$

So to use Shannon you must go backwards first:
S/N = 10 ^ (SNR(dB) / 10)

Worth knowing by heart

3 dB	10 dB	20 dB	25 dB	30 dB	40 dB
≈ 2	10	100	≈ 316	1,000	10,000

The classic exam error: putting 30 straight into Shannon's formula instead of 1,000. If your answer looks far too small, this is almost always why.

Remember from Part 1: dB is logarithmic, so every 10 dB is another factor of ten, and losses in dB simply add.

Shannon: the ceiling nothing beats

Claude Shannon, 1948 — the single most important result in this course

$$C = B \cdot \log_2 (1 + S/N) \quad \text{bit/s}$$

the maximum error-free rate of a channel of bandwidth B hertz carrying a signal-to-noise ratio S/N

B

The channel's bandwidth in hertz. Double it and you double the capacity — a straight, cheap win if you can get the spectrum.

S/N

Inside a logarithm. Doubling the transmit power buys you only about one extra bit per symbol. Power is an expensive way to go faster.

No M

There is no "number of levels" term. Shannon does not care how clever your encoding is — this is a limit of the channel itself.

No modulation, no coding, no compression and no future technology gets past this number. Beat it and you have broken physics, not engineering.

Shannon, worked

The same telephone line: B = 3,100 Hz, SNR = 30 dB

1	**Convert the decibels**	$SNR(dB) = 30$, so $S/N = 10^{(30/10)} = 10^3 = 1{,}000$	*the step people skip*
2	**Put it in the formula**	$C = 3{,}100 \times \log_2(1 + 1{,}000) = 3{,}100 \times \log_2(1{,}001)$	
3	**Work the logarithm**	$\log_2(1{,}001) \approx 9.97$	
4	**Multiply**	$C \approx 3{,}100 \times 9.97 \approx 30{,}900$ bit/s $\approx$ 30.9 kbit/s	*the answer*

Now put the two results together. Shannon says 30.9 kbit/s is the ceiling. Nyquist then tells you what it costs: 30,900 = 2 × 3,100 × $\log_2$ M gives $\log_2$ M ≈ 4.98, so about 32 levels. That is why a 33.6 k modem was the practical end of the line.

PAUSE POINT 5 · ABOUT 6 MINUTES

Your turn — three capacity problems

Pause here. Calculator out. Write the dB conversion down as a separate line every time.

1. **A channel has B = 1 MHz and an SNR of 20 dB. What is the Shannon capacity?** — Shannon

2. **A noiseless channel of 4 kHz carries 8 levels. What is the maximum bit rate?** — Nyquist

3. **A Wi-Fi channel is 80 MHz wide with an SNR of 25 dB. What is the capacity of one stream — and roughly what would two streams give?** — Shannon

A hint for question 3: the answer is going to look familiar. It is the number on the side of a Wi-Fi box.

Reminder: $\log_2 x = \log_{10} x \div \log_{10} 2 \approx \log_{10} x \div 0.301$, if your calculator has no log base 2 key.

ANSWERS · PAUSE POINT 5

Check your working

The dB conversion is its own line in every one of them.

1 B = 1 MHz, SNR 20 dB

$20\ \text{dB} \rightarrow S/N = 10^2 = 100$

$C = 1{,}000{,}000 \times \log_2(101)$

$\log_2(101) \approx 6.66$

$C \approx 6.66$ Mbit/s

2 B = 4 kHz, M = 8, noiseless

$\log_2 8 = 3$ bits per symbol

$C = 2 \times 4{,}000 \times 3$

$C = 24{,}000$ bit/s $= 24$ kbit/s

Nyquist: no SNR appears

3 B = 80 MHz, SNR 25 dB

$25\ \text{dB} \rightarrow S/N \approx 316$

$C = 80{,}000{,}000 \times \log_2(317)$

$\log_2(317) \approx 8.31$

$C \approx 665$ Mbit/s · two streams
≈ 1.3 Gbit/s

Question 3 is the answer to the Wi-Fi question from Class 3. 1200 Mbps on the box assumes a wide channel, two streams and a near-perfect signal. Move to the next room, the SNR falls, and Shannon pulls the ceiling down with it — before any of the Class 3 overheads are counted.

Which formula does the question want?

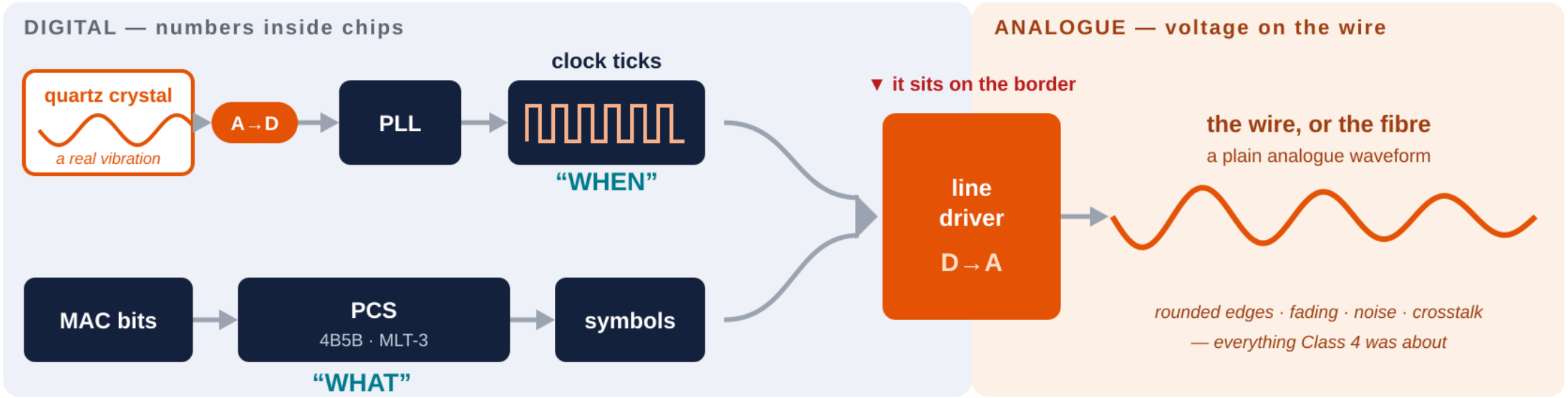
Read it for two words: “noiseless” and “levels”

	NYQUIST	SHANNON
What it assumes	A perfect channel with no noise at all	A real channel with noise
The formula	$C = 2B \cdot \log_2 M$	$C = B \cdot \log_2(1 + S/N)$
What you must be given	B and the number of levels M	B and the SNR (almost always in dB)
What it tells you	How fast your chosen encoding can run	The ceiling no encoding can beat
Give-away words	“noiseless”, “levels”, “signal elements”	“SNR”, “dB”, “noisy”, “capacity”
It is an	engineering answer	upper bound

If a question gives you both M and an SNR, work out both. The real rate is the smaller of the two — your encoding cannot run faster than Nyquist allows, and nothing can run faster than Shannon allows.

Where the digital world ends

Two streams meet at the line driver — and only one thing actually travels



And at the far end it happens in reverse: the receiver measures the voltage, decides which level was sent — A→D — and bits exist again.

Nothing on the wire is digital. "Digital" is a promise about how to read a voltage at each tick — and this whole class has been about what it costs to keep that promise.

The crystal is the metronome
It carries no data at all. It only says when.

The MAC is the music
The bits carry no timing. They only say what.

Neither works alone
Stop either one and the link stops.

Seven things to leave with

If you can say all seven in your own words, the video worked

1	**Layer 1 is the road**	It turns bits into voltage, light or radio, and knows nothing about frames or addresses.
2	**Two delays**	Transmission = L / R, propagation = d / s. A faster link only helps the first.
3	**Three impairments**	Attenuation, noise and distortion damage every signal. Losses in dB simply add.
4	**The clock problem**	The receiver needs a clock, and long runs of identical bits make it lose count.
5	**Line codes trade**	Manchester always changes but costs 100%; 4B5B costs 25%; 64b/66b costs 3%.
6	**Right medium, right job**	Copper for the last 100 m, fibre for distance, radio for freedom — physics sets each limit.
7	**Nyquist and Shannon**	Nyquist counts symbols on a clean channel; Shannon caps the bits once noise is in the picture.

THAT IS THE LAST VIDEO

Next: the last class, and it is in the room

Class 5 · LIVE · hands on

- Modulation: how bits ride on a wave — and how it cashes in the levels Nyquist allows
- Multiplexing: many signals on one medium, the sequel to Class 3’s channel partitioning

Then the whole course

- Layer 7 down to Layer 1 on one page
- The complete answer to the 1200 Mbps question
- What to revise, and what the exam actually asks

To think about before Class 5: Nyquist says more levels means more bits. Shannon says noise decides how many levels you can get away with. So how does a radio actually build those levels out of a wave?